

Appendices

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1 Detailed Derivation of the Local Lorentz Gauge Test, Torsion Invariants and Tetrad Diffeomorphism Gauge Test

In this appendix we provide the full derivation of the metric components, integrability conditions and the diffeomorphism gauge test associated with the tetrad ansatz used in the main text.

1.1 Necessary Mathematical Tools

1.1.1 Tetrad Ansatz

We begin with the tetrad field

$$e^A{}_\mu = \begin{pmatrix} c(t) & 0 & 0 & 0 \\ W_x & a(t) & 0 & 0 \\ W_y & 0 & a(t) & 0 \\ W_z & 0 & 0 & a(t) \end{pmatrix}, \quad (1)$$

where

$$W_x = b(t, x, y, z)x, \quad (2)$$

$$W_y = b(t, x, y, z)y, \quad (3)$$

$$W_z = b(t, x, y, z)z. \quad (4)$$

Equivalently,

$$W_i = b(t, \mathbf{x})x_i. \quad (5)$$

The tangent-space metric is chosen as

$$\eta_{AB} = \text{diag}(-1, +1, +1, +1). \quad (6)$$

1.1.2 Inverse Tetrad

The inverse tetrad satisfies

$$e^A{}_\mu e_B{}^\mu = \delta_B^A, \quad (7)$$

and

$$e^A{}_\mu e_A{}^\nu = \delta_\mu^\nu. \quad (8)$$

The tetrad matrix has the block form

$$e^A{}_\mu = \begin{pmatrix} c & 0 \\ W_i & a\delta_{ij} \end{pmatrix}. \quad (9)$$

Since the matrix is lower triangular, its inverse is

$$e_A{}^\mu = \begin{pmatrix} \frac{1}{c} & 0 \\ -\frac{1}{ac}W_i & \frac{1}{a}\delta_{ij} \end{pmatrix}. \quad (10)$$

Explicitly,

$$e_A{}^\mu = \begin{pmatrix} \frac{1}{c} & 0 & 0 & 0 \\ -\frac{W_x}{ac} & \frac{1}{a} & 0 & 0 \\ -\frac{W_y}{ac} & 0 & \frac{1}{a} & 0 \\ -\frac{W_z}{ac} & 0 & 0 & \frac{1}{a} \end{pmatrix}. \quad (11)$$

Substituting $W_i = bx_i$,

$$e_A{}^\mu = \begin{pmatrix} \frac{1}{c} & 0 & 0 & 0 \\ -\frac{bx}{ac} & \frac{1}{a} & 0 & 0 \\ -\frac{by}{ac} & 0 & \frac{1}{a} & 0 \\ -\frac{bz}{ac} & 0 & 0 & \frac{1}{a} \end{pmatrix}. \quad (12)$$

1.1.3 Metric Components

The metric is constructed in 2.

1.2 Local Lorentz Covariance Test of the Drift Sector

The remaining question is whether the drift field W_i can be removed by a local Lorentz transformation, specifically by a local boost. If this were possible, then the drift sector would represent only an inertial artifact rather than a genuine physical contribution.

This issue is particularly important in teleparallel gravity, because the tetrad contains both gravitational and inertial information. A quantity that can be eliminated by a local Lorentz transformation does not correspond to an independent physical degree of freedom.

The tetrad under consideration is

$$e^0 = c dt, \quad e^i = a(t) dx^i + W_i dt.$$

For simplicity we consider the static limit

$$a \rightarrow 1, \quad c \rightarrow 1,$$

so that

$$e^0 = dt, \quad e^i = dx^i + W_i dt.$$

A local Lorentz boost with velocity parameter $v_i(x)$ acts on the tetrad as

$$e'^a{}_\mu = \Lambda^a{}_b(x) e^b{}_\mu.$$

For infinitesimal boosts,

$$\Lambda^0{}_i = v_i, \quad \Lambda^i{}_0 = v_i,$$

and therefore the spatial tetrad transforms as

$$e'^i = e^i + v_i e^0.$$

Substituting the tetrad components gives

$$e'^i = dx^i + (W_i + v_i) dt.$$

Hence choosing

$$v_i = -W_i$$

formally removes the drift term from the tetrad:

$$e'^i = dx^i.$$

At first sight this suggests that W_i might be pure Lorentz gauge. However, in covariant teleparallel gravity the torsion tensor is not determined solely by the tetrad. One must simultaneously transform the spin connection according to

$$\omega'^a{}_{b\mu} = \Lambda^a{}_c \omega^c{}_{d\mu} (\Lambda^{-1})^d{}_b + \Lambda^a{}_c \partial_\mu (\Lambda^{-1})^c{}_b.$$

Even if the boosted tetrad becomes trivial, the transformed spin connection acquires inertial contributions proportional to derivatives of the boost parameter:

$$\partial_\mu v_i = -\partial_\mu W_i.$$

Therefore the torsion does not generally vanish. Instead, the torsion is redistributed

between the tetrad and the inertial spin connection.

The torsion tensor in covariant teleparallel gravity is

$$T^a_{\mu\nu} = \partial_\mu e^a_\nu - \partial_\nu e^a_\mu + \omega^a_{b\mu} e^b_\nu - \omega^a_{b\nu} e^b_\mu.$$

For the drift tetrad one obtains

$$T^i_{0j} = \partial_j W_i.$$

In the radial configuration

$$W_i = W(r)n_i,$$

this becomes

$$T^i_{0j} = \frac{W(r)}{r} (\delta_{ij} - n_i n_j) + W'(r) n_i n_j.$$

The associated torsion vector is

$$T_0 = W'(r) + \frac{2W(r)}{r}.$$

Thus the invariant

$$T_\mu T^\mu = - \left(W'(r) + \frac{2W(r)}{r} \right)^2$$

is nonzero whenever the drift profile is nontrivial.

This is crucial because $T_\mu T^\mu$ is a Lorentz scalar in the covariant formulation. A local Lorentz transformation may reshuffle torsion between the tetrad and the spin connection, but it cannot eliminate a nonzero invariant scalar.

Therefore, even if a local boost removes W_i from the explicit tetrad components, the physical torsion content remains encoded in the covariant torsion tensor.

Consequently,

$$W_i \text{ is not removable as a pure Lorentz gauge mode whenever } T_\mu T^\mu \neq 0.$$

This establishes the physical character of the drift sector in the GDV framework. The drift field is not merely a coordinate artifact nor a removable inertial mode; instead it represents a genuine torsional geometric structure contributing dynamically through the action

$$S = \frac{1}{2\kappa} \int d^4x e (T + \beta T_\mu T^\mu) + S_{\text{matter}}.$$

Hence for

$$\beta \neq 0,$$

the drift sector constitutes a true physical component of spacetime torsion.

Lorentz Covariance Test for the Tetrad

In this appendix, the lorentz covarience will be tested. To do that, consider the tetrad

$$e^0 = dt \quad e^i = a(t) dx^i + W_i dt,$$

where W

$$W_i = b(t, x) x_i.$$

Initially the Weitzenböck gauge,

$$\omega^a_b = 0.$$

The torsion 2-form is

$$T^a = de^a + \omega^a_b \wedge e^b.$$

Since initially

$$\omega^a_b = 0,$$

simply the torsion is

$$T^a = de^a.$$

Here the temporal and spatial parts of torsion must be considered seperately,

$$T^0 = de^0.$$

As $e^0 = dt$, the temporal component of the torsion is

$$T^0 = d(dt) = 0.$$

The spatial torsion is computed as follows,

$$T^i = de^i.$$

Using Eq. 1.2

$$de^i = d(a dx^i) + d(W_i dt).$$

There the components in the summation must be examined parts by parts.

Derivative of the FLRW Part First,

$$d(a dx^i) = da \wedge dx^i.$$

Since

$$da = \dot{a} dt,$$

this gives

$$d(a dx^i) = \dot{a} dt \wedge dx^i.$$

Derivative of the Shift Part

Now compute

$$d(W_i dt).$$

Applying the Leibniz rule,

$$d(W_i dt) = dW_i \wedge dt + W_i d(dt).$$

Since

$$d(W_i dt) = dW_i \wedge dt.$$

Now expand

$$dW_i.$$

Because $W_i = W_i(t, x)$,

$$dW_i = \partial_j W_i dx^j + \dot{W}_i dt.$$

Therefore,

$$dW_i \wedge dt = \partial_j W_i dx^j \wedge dt + \dot{W}_i dt \wedge dt.$$

Since

$$dt \wedge dt = 0,$$

we get

$$dW_i \wedge dt = \partial_j W_i dx^j \wedge dt.$$

Using antisymmetry,

$$dx^j \wedge dt = -dt \wedge dx^j,$$

hence

$$dW_i \wedge dt = -\partial_j W_i dt \wedge dx^j.$$

Spatial Torsion Substituting everything back,

$$T^i = \dot{a} dt \wedge dx^i - \partial_j W_i dt \wedge dx^j.$$

Rewrite the first term:

$$\dot{a} dt \wedge dx^i = \dot{a} \delta^i_j dt \wedge dx^j.$$

Thus,

$$T^i = (\dot{a} \delta^i_j - \partial_j W_i) dt \wedge dx^j.$$

Now by substituting Eq. 1.2,

$$\partial_j W_i = \partial_j (bx_i).$$

is obtained. Applying the product rule,

$$\partial_j (bx_i) = (\partial_j b)x_i + b \partial_j x_i.$$

Since

$$\partial_j x_i = \delta_{ij},$$

we obtain

$$\partial_j W_i = x_i \partial_j b + b \delta_{ij}.$$

Substituting into the torsion tensor,

$$T^i = (\dot{a} \delta^i_j - b \delta^i_j - x_i \partial_j b) dt \wedge dx^j.$$

Therefore,

$$T^i = [(\dot{a} - b) \delta^i_j - x_i \partial_j b] dt \wedge dx^j$$

This is the initial torsion.

Lorentz Boost

In this context, there are two ways to compute a Lorentz boost: the infinitesimal Lorentz boost and the general Lorentz boost. In the case of a general Lorentz boost, nonlinear terms appear. We now attempt to eliminate the shift vector W_i by means of a local Lorentz boost.

Infinitesimal Local Lorentz Boost Choose the infinitesimal boost parameter

$$\beta_i = -W_i = -bx_i$$

The infinitesimal Lorentz matrix is

$$\Lambda^0_0 = 1,$$

$$\Lambda^0_i = \beta_i,$$

$$\Lambda^i_0 = \beta_i,$$

$$\Lambda^i_j = \delta^i_j.$$

The transformed tetrad is

$$e'^a = \Lambda^a_b e^b.$$

Again, the spatal and temporal tetrads are examined individually. In this case firstly the spatial tetrad is,

$$e'^i.$$

Using

$$e'^i = \Lambda^i_0 e^0 + \Lambda^i_j e^j,$$

the following form is obtained,

$$e'^i = \beta_i e^0 + \delta^i_j e^j.$$

By substituting Eq.1.2, it gives

$$e'^i = \beta_i dt + a dx^i + W_i dt.$$

Using Eq. 1.2, the following equation is obtained

$$e'^i = -W_i dt + a dx^i + W_i dt.$$

Hence

$$e'^i = a dx^i$$

The shift vector disappears from the spatial tetrad. Now, the temporal tetrad is

$$e'^0.$$

Using

$$e'^0 = \Lambda^0_0 e^0 + \Lambda^0_i e^i,$$

we obtain

$$e'^0 = dt + \beta_i (a dx^i + W_i dt).$$

Expanding,

$$e'^0 = dt + a\beta_i dx^i + \beta_i W_i dt.$$

The last term is quadratic:

$$\beta_i W_i \sim W^2.$$

Keeping only linear order, due to the infinitesimal Lorentz Boost

$$e'^0 = dt + a\beta_i dx^i.$$

By using Eq. 1.2,

$$e'^0 = dt - aW_i dx^i$$

is acquired. Thus the Lorentz transformation does not leave e^0 unchanged.

Generated Spin Connection Initially,

$$\omega^a_b = 0.$$

After a local Lorentz transformation,

$$\omega'^a_b = \Lambda^a_c \omega^c_d (\Lambda^{-1})^d_b + \Lambda^a_c d(\Lambda^{-1})^c_b.$$

Since initially Weitzenböck Connection is used and it is obtain

$$\omega'^a_b = \Lambda^a_c d(\Lambda^{-1})^c_b.$$

For an infinitesimal transformation,

$$\Lambda^a_b = \delta^a_b + \epsilon^a_b.$$

Then

$$(\Lambda^{-1})^a_b = \delta^a_b - \epsilon^a_b.$$

Hence

$$d(\Lambda^{-1})^a_b = -d\epsilon^a_b.$$

Substituting,

$$\omega'^a_b = -d\epsilon^a_b.$$

Because

$$\epsilon^i_0 = \beta_i,$$

we obtain

$$\omega'^i_0 = -d\beta_i.$$

Using

$$\beta_i = -W_i,$$

we finally get

$$\boxed{\omega'^i_0 = dW_i}$$

Now compute dW_i explicitly.

Since

$$W_i = bx_i,$$

we have

$$dW_i = d(bx_i).$$

Applying the product rule,

$$dW_i = x_i db + b dx_i.$$

Now compute

$$db.$$

Because $b = b(t, x)$,

$$db = \dot{b} dt + \partial_j b dx^j.$$

Therefore,

$$dW_i = x_i \dot{b} dt + x_i \partial_j b dx^j + b dx_i.$$

Hence,

$$\boxed{\omega^i{}_0 = x_i \dot{b} dt + x_i \partial_j b dx^j + b dx_i}$$

Transformed Spatial Torsion

Now compute

$$T'^i = de'^i + \omega^i{}_0 \wedge e'^0.$$

Derivative Term

- Since

$$e'^i = a dx^i,$$

we obtain

$$de'^i = d(a dx^i) = da \wedge dx^i.$$

Using

$$da = \dot{a} dt,$$

gives

$$de'^i = \dot{a} dt \wedge dx^i.$$

Connection Term Now compute

$$\omega^i{}_0 \wedge e'^0.$$

Using

$$e'^0 = dt - a W_k dx^k,$$

we obtain

$$\omega'^i{}_0 \wedge e'^0 = dW_i \wedge (dt - aW_k dx^k).$$

Expanding,

$$\omega'^i{}_0 \wedge e'^0 = dW_i \wedge dt - aW_k dW_i \wedge dx^k.$$

The second term is quadratic:

$$W_k dW_i \sim W^2.$$

At linear order,

$$\omega'^i{}_0 \wedge e'^0 \simeq dW_i \wedge dt.$$

Now substitute

$$dW_i = \partial_j W_i dx^j + \dot{W}_i dt.$$

Then

$$dW_i \wedge dt = \partial_j W_i dx^j \wedge dt.$$

Using antisymmetry,

$$dx^j \wedge dt = -dt \wedge dx^j,$$

we get

$$dW_i \wedge dt = -\partial_j W_i dt \wedge dx^j.$$

Result for T'^i Therefore,

$$T'^i = \dot{a} dt \wedge dx^i - \partial_j W_i dt \wedge dx^j.$$

Rewriting,

$$T'^i = (\dot{a}\delta^i{}_j - \partial_j W_i) dt \wedge dx^j.$$

Substituting

$$\partial_j W_i = b\delta_{ij} + x_i \partial_j b,$$

we obtain

$$\boxed{T'^i = [(\dot{a} - b)\delta^i{}_j - x_i \partial_j b] dt \wedge dx^j}$$

Thus,

$$\boxed{T'^i = T^i}$$

at linear order. However, when the non linear terms are not eliminated then, the relation is not satisfied.

Transformed Temporal Torsion

Now compute the temporal component.

The transformed torsion is

$$T'^0 = de'^0 + \omega'^0{}_i \wedge e'^i.$$

Derivative of e'^0

Using

$$e'^0 = dt - aW_i dx^i,$$

we obtain

$$de'^0 = -d(aW_i dx^i).$$

Applying the Leibniz rule,

$$de'^0 = -d(aW_i) \wedge dx^i - aW_i d(dx^i).$$

Since

$$d(dx^i) = 0,$$

we get

$$de'^0 = -d(aW_i) \wedge dx^i.$$

Now compute

$$d(aW_i).$$

Using the product rule,

$$d(aW_i) = da W_i + a dW_i.$$

Since

$$da = \dot{a} dt,$$

we obtain

$$d(aW_i) = \dot{a}W_i dt + a dW_i.$$

Therefore,

$$de'^0 = -\dot{a}W_i dt \wedge dx^i - a dW_i \wedge dx^i.$$

Connection Contribution Now compute

$$\omega'^0{}_i \wedge e'^i.$$

Using

$$\omega'^0{}_i = dW_i,$$

and

$$e'^i = a dx^i,$$

we obtain

$$\omega'^0{}_i \wedge e'^i = a dW_i \wedge dx^i.$$

Result for T'^0

Adding both terms,

$$T'^0 = -\dot{a}W_i dt \wedge dx^i - a dW_i \wedge dx^i + a dW_i \wedge dx^i.$$

The spatial two-form terms cancel:

$$-a dW_i \wedge dx^i + a dW_i \wedge dx^i = 0.$$

Hence,

$$\boxed{T'^0 = -\dot{a}W_i dt \wedge dx^i}$$

In general,

$$\boxed{T'^0 \neq 0}$$

whereas initially

$$\boxed{T^0 = 0}$$

Therefore, the full test's results is

$$\boxed{T'^a \neq T^a}$$

component-wise.

1.3 Detailed Derivation of the Torsion Invariants Before and After Local Lorentz Boosts

In this appendix we derive explicitly the quadratic torsion invariants, the torsion vector norm, the axial torsion norm, and the TEGR torsion scalar both before and after the local Lorentz boost.

Tangent-Space Torsion Tensor

The tangent-space torsion tensor is defined by

$$T_{ABC} = \eta_{AD} T^D{}_{BC}, \tag{13}$$

with

$$T^A{}_{BC} = E^\mu{}_B E^\nu{}_C (\partial_\mu e^A{}_\nu - \partial_\nu e^A{}_\mu). \quad (14)$$

For the tetrad

$$e^A{}_\mu = \begin{pmatrix} c & 0 & 0 & 0 \\ bx & a & 0 & 0 \\ by & 0 & a & 0 \\ bz & 0 & 0 & a \end{pmatrix}, \quad (15)$$

the nonvanishing tangent-space torsion components are

$$T_{101} = \frac{-x\partial_x b - b + \dot{a}}{ac}, \quad (16)$$

$$T_{102} = -\frac{x\partial_y b}{ac}, \quad (17)$$

$$T_{103} = -\frac{x\partial_z b}{ac}, \quad (18)$$

$$T_{202} = \frac{-y\partial_y b - b + \dot{a}}{ac}, \quad (19)$$

$$T_{203} = -\frac{y\partial_z b}{ac}, \quad (20)$$

$$T_{303} = \frac{-z\partial_z b - b + \dot{a}}{ac}. \quad (21)$$

Using antisymmetry in the last two indices,

$$T_{ABC} = -T_{ACB}, \quad (22)$$

the remaining components follow automatically.

First Quadratic Invariant I_1 The first invariant is

$$I_1 = T_{ABC} T^{ABC}. \quad (23)$$

Raising indices gives

$$T^{ABC} = \eta^{AA'} \eta^{BB'} \eta^{CC'} T_{A'B'C'}. \quad (24)$$

Since

$$\eta_{AB} = \text{diag}(-1, 1, 1, 1), \quad (25)$$

every temporal index contributes one minus sign.

Contribution from T_{101}

We compute

$$T_{101}T^{101} = \eta^{11}\eta^{00}\eta^{11}(T_{101})^2 \quad (26)$$

$$= (+1)(-1)(+1)(T_{101})^2 \quad (27)$$

$$= -(T_{101})^2. \quad (28)$$

Substituting,

$$T_{101}T^{101} = - \left(\frac{x\partial_x b + b - \dot{a}}{ac} \right)^2. \quad (29)$$

Contribution from T_{102}

Similarly,

$$T_{102}T^{102} = \eta^{11}\eta^{00}\eta^{22}(T_{102})^2 \quad (30)$$

$$= -(T_{102})^2 \quad (31)$$

$$= - \left(\frac{x\partial_y b}{ac} \right)^2. \quad (32)$$

Likewise,

$$T_{103}T^{103} = - \left(\frac{x\partial_z b}{ac} \right)^2, \quad (33)$$

$$T_{202}T^{202} = - \left(\frac{y\partial_y b + b - \dot{a}}{ac} \right)^2, \quad (34)$$

$$T_{203}T^{203} = - \left(\frac{y\partial_z b}{ac} \right)^2, \quad (35)$$

$$T_{303}T^{303} = - \left(\frac{z\partial_z b + b - \dot{a}}{ac} \right)^2. \quad (36)$$

Full Expression for I_1 Collecting all terms,

$$\begin{aligned}
I_1 = -\frac{2}{a^2 c^2} & \left[x^2 (\partial_y b)^2 + x^2 (\partial_z b)^2 + y^2 (\partial_x b)^2 \right. \\
& + y^2 (\partial_z b)^2 + z^2 (\partial_x b)^2 + z^2 (\partial_y b)^2 \\
& + (x \partial_x b + b - \dot{a})^2 \\
& + (y \partial_y b + b - \dot{a})^2 \\
& \left. + (z \partial_z b + b - \dot{a})^2 \right]. \tag{37}
\end{aligned}$$

The second invariant is

$$I_2 = T_{ABC} T^{CBA}. \tag{38}$$

Sample Calculation

Consider the contribution from T_{102} :

$$T_{102} T^{201} = T_{102} \eta^{22} \eta^{00} \eta^{11} T_{201} \tag{39}$$

$$= -T_{102} T_{201}. \tag{40}$$

Using

$$T_{102} = -\frac{x \partial_y b}{ac}, \tag{41}$$

and

$$T_{201} = -\frac{y \partial_x b}{ac}, \tag{42}$$

we obtain

$$T_{102} T^{201} = - \left(-\frac{x \partial_y b}{ac} \right) \left(-\frac{y \partial_x b}{ac} \right) \tag{43}$$

$$= -\frac{xy (\partial_x b) (\partial_y b)}{a^2 c^2}. \tag{44}$$

Complete Expression for I_2

Collecting all terms,

$$\begin{aligned}
I_2 = -\frac{1}{a^2 c^2} & \left[2xy(\partial_x b)(\partial_y b) + 2xz(\partial_x b)(\partial_z b) \right. \\
& + 2yz(\partial_y b)(\partial_z b) \\
& + (x\partial_x b + b - \dot{a})^2 \\
& + (y\partial_y b + b - \dot{a})^2 \\
& \left. + (z\partial_z b + b - \dot{a})^2 \right]. \tag{45}
\end{aligned}$$

Torsion Vector The torsion vector is

$$T_A = T^B{}_{BA}. \tag{46}$$

Temporal Component We compute

$$T_0 = T^1{}_{10} + T^2{}_{20} + T^3{}_{30}. \tag{47}$$

Substituting,

$$\begin{aligned}
T_0 = & \frac{x\partial_x b + b - \dot{a}}{ac} + \frac{y\partial_y b + b - \dot{a}}{ac} \\
& + \frac{z\partial_z b + b - \dot{a}}{ac}. \tag{48}
\end{aligned}$$

Thus,

$$T_0 = \frac{x\partial_x b + y\partial_y b + z\partial_z b + 3b - 3\dot{a}}{ac}. \tag{49}$$

The spatial components vanish,

$$T_1 = T_2 = T_3 = 0. \tag{50}$$

Norm of the Torsion Vector The norm is

$$T_A T^A = \eta^{AB} T_A T_B. \tag{51}$$

Since only $T_0 \neq 0$,

$$T_A T^A = \eta^{00} (T_0)^2 \tag{52}$$

$$= -(T_0)^2. \tag{53}$$

Therefore,

$$T_A T^A = -\frac{(x\partial_x b + y\partial_y b + z\partial_z b + 3b - 3\dot{a})^2}{a^2 c^2}. \quad (54)$$

Axial Vector The axial vector is defined as

$$A^A = \frac{1}{6}\epsilon^{ABCD}T_{BCD}. \quad (55)$$

Computation of A^1 We compute

$$A^1 = \frac{1}{6}\epsilon^{1BCD}T_{BCD}. \quad (56)$$

The nonvanishing contributions are

$$A^1 = \frac{1}{6}(\epsilon^{1023}T_{023} + \epsilon^{1032}T_{032}). \quad (57)$$

Using

$$\epsilon^{1023} = +1, \quad \epsilon^{1032} = -1, \quad (58)$$

we obtain

$$A^1 = \frac{1}{6}(T_{023} - T_{032}). \quad (59)$$

Substituting the torsion components,

$$A^1 = \frac{y\partial_z b - z\partial_y b}{3ac}. \quad (60)$$

Similarly,

$$A^2 = \frac{-z\partial_x b + x\partial_z b}{3ac}, \quad (61)$$

$$A^3 = \frac{x\partial_y b - y\partial_x b}{3ac}. \quad (62)$$

Norm of the Axial Vector

The norm is

$$A_A A^A = \eta_{AB} A^A A^B. \quad (63)$$

Since $A^0 = 0$,

$$A_A A^A = (A^1)^2 + (A^2)^2 + (A^3)^2. \quad (64)$$

Therefore,

$$A_A A^A = \frac{(x\partial_y b - y\partial_x b)^2 + (x\partial_z b - z\partial_x b)^2 + (y\partial_z b - z\partial_y b)^2}{9a^2 c^2}. \quad (65)$$

TEGR Torsion Scalar The TEGR scalar is

$$T = \frac{1}{4}I_1 + \frac{1}{2}I_2 - T_A T^A. \quad (66)$$

Substituting the explicit expressions for I_1 , I_2 , and $T_A T^A$ yields the full scalar reported in the symbolic computation.

Local Lorentz Boost

We now perform a local Lorentz boost in the x -direction,

$$\Lambda^A{}_B = \begin{pmatrix} \gamma & -\gamma\beta & 0 & 0 \\ -\gamma\beta & \gamma & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}, \quad (67)$$

with

$$\gamma = \frac{1}{\sqrt{1 - \beta^2}}. \quad (68)$$

The boosted torsion tensor becomes

$$T'^A{}_{BC} = \Lambda^A{}_D (\Lambda^{-1})^E{}_B (\Lambda^{-1})^F{}_C T^D{}_{EF}. \quad (69)$$

Boosted Torsion Vector Under the boost,

$$T'_0 = \gamma T_0, \quad (70)$$

$$T'_1 = \gamma\beta T_0, \quad (71)$$

$$T'_2 = 0, \quad (72)$$

$$T'_3 = 0. \quad (73)$$

Boosted Vector Norm We compute

$$T'_A T'^A = -(T'_0)^2 + (T'_1)^2 \quad (74)$$

$$= -\gamma^2 T_0^2 + \gamma^2 \beta^2 T_0^2 \quad (75)$$

$$= -\gamma^2 (1 - \beta^2) T_0^2. \quad (76)$$

Using

$$\gamma^2 = \frac{1}{1 - \beta^2}, \quad (77)$$

we obtain

$$T'_A T'^A = -T_0^2 \quad (78)$$

$$= T_A T^A. \quad (79)$$

Thus the torsion vector norm is invariant.

Boost Invariance of the Scalars

The symbolic computation gives

$$I'_1 - I_1 = 0, \quad (80)$$

$$I'_2 - I_2 = 0, \quad (81)$$

$$T'_A T'^A - T_A T^A = 0, \quad (82)$$

$$A'_A A'^A - A_A A^A = 0, \quad (83)$$

and finally

$$T' - T = 0. \quad (84)$$

Hence all quadratic torsion invariants remain unchanged under the local Lorentz boost when the corresponding spin connection is properly included.

2 Derivation of the Metric

$$\eta_{ab} = \text{diag}(-1, 1, 1, 1), \quad (85)$$

$$ds^2 = \eta_{ab} e^a e^b = -(e^0)^2 + \delta_{ij} e^i e^j. \quad (86)$$

$$ds^2 = -c^2 dt^2 + \delta_{ij} (a dx^i + W_i dt) (a dx^j + W_j dt). \quad (87)$$

$$ds^2 = -(c^2 - W^2) dt^2 + 2a W_i dx^i dt + a^2 \delta_{ij} dx^i dx^j, \quad W^2 := \delta_{ij} W^i W^j. \quad (88)$$

$$g_{tt} = -c^2 + W^2, \quad g_{ti} = a W_i, \quad g_{ij} = a^2 \delta_{ij}. \quad (89)$$

$$N = c, \quad N_i = a W_i, \quad \gamma_{ij} = a^2 \delta_{ij}. \quad (90)$$

$$g^{tt} = -\frac{1}{c^2}, \quad g^{ti} = \frac{W^i}{a c^2}, \quad g^{ij} = \frac{1}{a^2} \delta^{ij} - \frac{W^i W^j}{a^2 c^2}. \quad (91)$$

The matrix form is,

$$g_{\mu\nu} = \begin{pmatrix} r^2 b^2(t, r) - c^2(t) & x a(t) b(t, r) & y a(t) b(t, r) & z a(t) b(t, r) \\ x a(t) b(t, r) & a^2(t) & 0 & 0 \\ y a(t) b(t, r) & 0 & a^2(t) & 0 \\ z a(t) b(t, r) & 0 & 0 & a^2(t) \end{pmatrix}$$

where ,

$$r = \sqrt{x^2 + y^2 + z^2}.$$

3 Torsion Derivations for Various Observers

3.1 Derivation of Torsion Tensor for the observer at rest

$$\omega^a_{b\mu} = 0, \quad T^a_{\mu\nu} = \partial_\mu e^a_\nu - \partial_\nu e^a_\mu, \quad T^\rho_{\mu\nu} = e_a^\rho T^a_{\mu\nu}. \quad (92)$$

$$e^0_0 = 1, \quad e^0_j = 0, \quad e^i_0 = W^i(t, \vec{x}), \quad e^i_j = a(t) \delta^i_j. \quad (93)$$

$$e_0^0 = 1, \quad e_0^j = -\frac{W^j}{a}, \quad e_i^0 = 0, \quad e_i^j = \frac{1}{a} \delta_i^j. \quad (94)$$

$$T^i_{0j} = e_a^i (\partial_0 e^a_j - \partial_j e^a_0) = e_k^i (\partial_0 e^k_j - \partial_j e^k_0) \quad (95)$$

$$= \frac{1}{a} \delta_k^i (\partial_0 (a \delta^k_j) - \partial_j W^k) = \frac{1}{a} (\dot{a} \delta^i_j - \partial_j W^i). \quad (96)$$

$$W^i(t, \vec{x}) = b(t) x^i - W^{\text{adapt } i}(t, \vec{x}) \implies \partial_j W^i = b(t) \delta^i_j - \partial_j W^{\text{adapt } i}. \quad (97)$$

$$T^i_{0j} = \frac{1}{a} [(\dot{a} - b) \delta^i_j + \partial_j W^{\text{adapt } i}]. \quad (98)$$

$$(T^i{}_{0j})_{\text{rest}} = \frac{1}{a} \left[(\dot{a} - b) \delta^i{}_j + \partial_j W^{\text{adapt } i} \right] \neq 0. \quad (99)$$

3.2 Derivation of Torsion Tensor for freely falling observer

$$\omega^a{}_{b\mu} = 0, \quad T^a{}_{\mu\nu} = \partial_\mu e^a{}_\nu - \partial_\nu e^a{}_\mu, \quad T^\rho{}_{\mu\nu} = e_a{}^\rho T^a{}_{\mu\nu}. \quad (100)$$

$$e^0{}_0 = 1, \quad e^0{}_j = 0, \quad e^i{}_0 = W^i(t, \vec{x}), \quad e^i{}_j = a(t) \delta^i{}_j. \quad (101)$$

$$e_0^0 = 1, \quad e_0^j = -\frac{W^j}{a}, \quad e_i^0 = 0, \quad e_i^j = \frac{1}{a} \delta_i^j. \quad (102)$$

$$u^\mu = \frac{dx^\mu}{dt}, \quad u^0 = 1, \quad v^i := a(t) u^i + W^i(t, \vec{x}). \quad (103)$$

$$\frac{dv^i}{dt} = \frac{d}{dt} (a u^i + W^i) = \dot{a} u^i + a \dot{u}^i + \partial_t W^i + u^k \partial_k W^i. \quad (104)$$

$$v^i = 0 \implies a u^i = -W^i. \quad (105)$$

$$\dot{a} u^i = -\frac{\dot{a}}{a} W^i. \quad (106)$$

$$(T^i{}_{0j})_{\text{free fall}} = e_a{}^i (\partial_0 e^a{}_j - \partial_j e^a{}_0) = e_k{}^i (\partial_0 e^k{}_j - \partial_j e^k{}_0). \quad (107)$$

$$= \frac{1}{a} \delta_k{}^i (\partial_0 (a \delta^k{}_j) - \partial_j W^k) = \frac{1}{a} (\dot{a} \delta^i{}_j - \partial_j W^i). \quad (108)$$

$$W^i = b(t) x^i - W^{\text{adapt } i} \implies \partial_j W^i = b(t) \delta^i{}_j - \partial_j W^{\text{adapt } i}. \quad (109)$$

$$(T^i{}_{0j})_{\text{free fall}} = \frac{1}{a} \left[(\dot{a} - b) \delta^i{}_j + \partial_j W^{\text{adapt } i} \right]. \quad (110)$$

4 Derivation of Contortion and Superpotential Tensors

4.1 Derivation of Contortion Tensors

For the tetrad structure employed in this work, the non-vanishing components of the contortion tensor are obtained as follows.

- The symmetric contortion component K^i_{0j} encodes the mismatch between the inertial flow of spacetime and the motion of a particle. This mismatch is determined by the geometric drift field W^i and represents the teleparallel geometric expression of inertial incompatibility.

$$K^{\mu\nu}{}_{\rho} = -\frac{1}{2} (T^{\mu\nu}{}_{\rho} - T^{\nu\mu}{}_{\rho} - T_{\rho}{}^{\mu\nu}) \quad (111)$$

$$K^i{}_{0j} = \frac{1}{2} (T^j{}_{i0} + T^i{}_{j0}), \quad (112)$$

- This antisymmetric component represents the local rotational or vortical structure of spacetime. Within the framework considered here, this term corresponds to the rotational contributions of the flow field and vanishes at cosmological scales due to isotropy, while producing Coriolis-like effects in local systems.

$$\begin{aligned} K^{i0}{}_j &= \frac{-1}{2} (T^{i0}{}_j - 0 - T_j{}^{i0}) \\ &= \frac{1}{2} (-T^{i0}{}_j + T_j{}^{i0}) \\ &= \frac{1}{2} (T^{ij}{}_0 + T_j{}^{i0}) \end{aligned} \quad (113)$$

- The algebraic relation between the component $K^0{}_{ij}$ and $K^i{}_{0j}$ indicates that the mismatch between temporal and spatial components originates from the same geometric source.

$$\begin{aligned} K^{ij}{}_0 &= \frac{-1}{2} (T^{ij}{}_0 - T^{ji}{}_0 - 0) \\ &= \frac{1}{2} (T^{ji}{}_0 - T^{ij}{}_0) \end{aligned} \quad (114)$$

- In contrast, the vanishing of the purely spatial components $K^i{}_{jk} = 0$ implies that space itself does not contain any intrinsic inertial distortion, shear, or rotation. All physical effects arise from the interplay between the temporal evolution of space and the underlying flow structure.

$$K^i{}_{jk} = 0. \quad (115)$$

Taken together, these results demonstrate that the contortion tensor carries physical content exclusively through time-space mixing components.

4.2 Derivation of Superpotential Tensors

$$S_{\rho}{}^{\mu\nu} = \frac{1}{2} (K^{\mu\nu}{}_{\rho} + \delta^{\mu}{}_{\rho} T^{\alpha\nu}{}_{\alpha} - \delta^{\nu}{}_{\rho} T^{\alpha\mu}{}_{\alpha}) \quad (116)$$

Superpotential components with purely spatial lower indices

$$S^0_{ij} = \frac{1}{2} K_{ij}{}^0 \quad (117)$$

are obtained.

$$\begin{aligned} S_i{}^{0j} &= \frac{1}{2}(K^{0j}{}_i + 0 - \delta^j_i T^{\alpha 0}{}_\alpha) \\ &= \frac{1}{2}(K^{0j}{}_i + 0 - \delta^j_i T_0) \\ &= \frac{-1}{2}[K^{j0}{}_i + \delta^j_i T_0] \\ &= \frac{-1}{2}[K^{i0}{}_j + \delta^i_j T_0] \end{aligned} \quad (118)$$

Although these components are associated with temporal energy fluxes induced by spatial distortions, they do not contribute to an independent gravitational energy transport within the drift-adapted frame and under the assumed symmetries.

- Time-space mixed components of the superpotential

$$S^i{}_{0j} = -\frac{1}{2} (K^i{}_{0j} + \delta^i_j T_0), \quad T_0 := T^\mu{}_{0\mu} \quad (119)$$

are obtained.

This term contains contributions from both the contortion tensor and the torsion trace. In this way, it combines the inertial structure of spacetime that is incompatible with free particle motion with the effects of global expansion and geometric drift, thereby representing the geometric origin of gravitational energy.

- Purely spatial superpotential components

$$S^i{}_{jk} = 0 \quad (120)$$

vanish identically.

5 Proof that $T_\mu T^\mu$ and Riemannian Geometry

Teleparallel gravity is related to the curvature formulation through the well-known identity

$$R = -T + 2\nabla_\mu T^\mu. \quad (121)$$

Equivalently,

$$T = -R + 2\nabla_\mu T^\mu. \quad (122)$$

Therefore, the TEGR action

$$S_{\text{TEGR}} = \frac{1}{2\kappa} \int d^4x e T \quad (123)$$

can be rewritten as

$$S_{\text{TEGR}} = \frac{1}{2\kappa} \int d^4x e (-R + 2\nabla_\mu T^\mu). \quad (124)$$

The second term is a total divergence:

$$\int d^4x e \nabla_\mu T^\mu. \quad (125)$$

Hence, under the usual boundary assumptions, it does not contribute to the bulk field equations. One obtains

$$S_{\text{TEGR}} \sim -\frac{1}{2\kappa} \int d^4x e R. \quad (126)$$

This is the standard equivalence between TEGR and GR. The equivalence works because the difference between the torsion scalar and the Ricci scalar is only a boundary term.

However, the theory considered in this work is not TEGR. Its gravitational action is

$$S = \frac{1}{2\kappa} \int d^4x e (T + \beta T_\mu T^\mu). \quad (127)$$

Using the same identity, this becomes

$$S = \frac{1}{2\kappa} \int d^4x e (-R + 2\nabla_\mu T^\mu + \beta T_\mu T^\mu). \quad (128)$$

Dropping the total divergence gives

$$S \sim -\frac{1}{2\kappa} \int d^4x e R + \frac{\beta}{2\kappa} \int d^4x e T_\mu T^\mu. \quad (129)$$

Thus the theory is not simply GR written in tetrad variables. It is GR plus an additional bulk torsion-vector contribution. The crucial point is that

$$T_\mu T^\mu \quad (130)$$

is not a boundary term. It contributes directly to the Euler–Lagrange equations.

This also shows why the torsion-vector sector cannot be reconstructed from the boundary scalar alone. Define

$$\nabla_\mu T^\mu = B. \quad (131)$$

The scalar B only fixes the divergence of the vector T^μ . It does not determine the vector itself. Indeed, one may shift

$$T^\mu \rightarrow T^\mu + V^\mu \quad (132)$$

with

$$\nabla_\mu V^\mu = 0. \quad (133)$$

Then

$$\nabla_\mu (T^\mu + V^\mu) = B. \quad (134)$$

Therefore, infinitely many torsion vectors have the same boundary scalar B . However, their quadratic norm is different:

$$(T_\mu + V_\mu)(T^\mu + V^\mu) = T_\mu T^\mu + 2T_\mu V^\mu + V_\mu V^\mu. \quad (135)$$

Hence $T_\mu T^\mu$ contains information that is not contained in $\nabla_\mu T^\mu$, and therefore not contained in the Ricci scalar R alone.

Consequently, the $\beta T_\mu T^\mu$ deformation cannot be absorbed into the Einstein–Hilbert action by integration by parts. It represents a genuinely new geometric sector. In the present model, this is precisely the sector in which the drift field W_i contributes.

Therefore,

$$\boxed{\beta T_\mu T^\mu \text{ is a bulk torsional invariant and cannot be rewritten purely as GR.}} \quad (136)$$

Equivalently,

$$\text{TEGR is equivalent to GR, but the present } T + \beta T_\mu T^\mu \quad (137)$$

6 Irreducible Torsion Invariants in the Drift Sector

The torsion tensor admits the irreducible decomposition

$$T_{\lambda\mu\nu} = \frac{1}{3} (g_{\lambda\nu} T_\mu - g_{\lambda\mu} T_\nu) + \frac{1}{6} \epsilon_{\lambda\mu\nu\rho} A^\rho + q_{\lambda\mu\nu}. \quad (138)$$

Quadratic invariants:

- Trace sector:

$$I_T = T_\mu T^\mu \quad (139)$$

- Axial sector:

$$I_A = A_\mu A^\mu, \quad A^\rho = \epsilon^{\rho\alpha\beta\gamma} T_{\alpha\beta\gamma} \quad (140)$$

- Tensor sector:

$$I_q = q_{\lambda\mu\nu} q^{\lambda\mu\nu} \quad (141)$$

Drift tetrad:

$$e^0 = dt, \quad e^i = a(t)dx^i + W^i dt, \quad W^i = b(t, x)x^i \quad (142)$$

Torsion:

$$T^i{}_{0j} = \frac{1}{a} (\dot{a} \delta^i{}_j - \partial_j W^i) \quad (143)$$

$$\partial_j W^i = b \delta^i{}_j + x^i \partial_j b \quad (144)$$

Homogeneous limit:

$$\partial_j b = 0 \quad (145)$$

$$T^i{}_{0j} = X \delta^i{}_j, \quad X := H - \frac{b}{a} \quad (146)$$

All other components vanish:

$$T^i{}_{jk} = 0, \quad T^0{}_{\mu\nu} = 0 \quad (147)$$

Trace invariant:

$$T_0 = T^\nu{}_{\nu 0} = -3X, \quad T_i = 0 \quad (148)$$

$$I_T = T_\mu T^\mu = -9X^2 \neq 0 \quad (149)$$

Axial invariant:

$$A^\rho = \epsilon^{\rho\alpha\beta\gamma} T_{\alpha\beta\gamma} \quad (150)$$

Only non-zero torsion components:

$$T_{\alpha\beta\gamma} \sim T_{i0j} \propto \delta_{ij} \quad (151)$$

Thus

$$\epsilon^{\rho i 0 j} \delta_{ij} = 0 \quad (152)$$

$$A^\rho = 0 \implies I_A = 0 \quad (153)$$

Tensor invariant:

$$q_{\lambda\mu\nu} = T_{\lambda\mu\nu} - \frac{1}{3}(g_{\lambda\nu}T_\mu - g_{\lambda\mu}T_\nu) \quad (154)$$

Substituting $T^i_{0j} = X\delta^i_j$:

$$q_{\lambda\mu\nu} = 0 \quad (155)$$

$$I_q = 0 \quad (156)$$

$$I_T \neq 0, \quad I_A = 0, \quad I_q = 0 \quad (157)$$

$$\text{Only } T_\mu T^\mu \text{ survives in the isotropic drift sector} \quad (158)$$

Action Solution: Action:

$$S = \frac{1}{2\kappa} \int d^4x e (T + \beta T_\mu T^\mu) \quad (159)$$

General quadratic extension:

$$S = \frac{1}{2\kappa} \int d^4x e (T + \alpha_1 T_\mu T^\mu + \alpha_2 A_\mu A^\mu + \alpha_3 q_{\lambda\mu\nu} q^{\lambda\mu\nu}) \quad (160)$$

Drift sector reduction:

$$A_\mu = 0, \quad q_{\lambda\mu\nu} = 0 \quad (161)$$

Thus

$$A_\mu A^\mu = 0, \quad q_{\lambda\mu\nu} q^{\lambda\mu\nu} = 0 \quad (162)$$

Action reduces to

$$S = \frac{1}{2\kappa} \int d^4x e (T + \alpha_1 T_\mu T^\mu) \quad (163)$$

$$\alpha_2, \alpha_3 \text{ drop out identically} \quad (164)$$

Variation:

$$\delta S = \frac{1}{2\kappa} \int d^4x e [E_a{}^\nu + \alpha_1 Z_a{}^\nu] \delta e^a{}_\nu \quad (165)$$

with

$$\delta(T_\mu T^\mu) = 2T^\mu \delta T_\mu + (T_\alpha T^\alpha) e_a{}^\nu \delta e^a{}_\nu \quad (166)$$

No contribution from axial/tensor sectors:

$$\delta(A_\mu A^\mu) = 0, \quad \delta(q^2) = 0 \quad (167)$$

EFT interpretation:

Operator expansion:

$$\mathcal{L} = T + \frac{1}{\Lambda^2} (c_1 T_\mu T^\mu + c_2 A_\mu A^\mu + c_3 q^2) + \mathcal{O}(\partial^4) \quad (168)$$

Drift background:

$$A_\mu = 0, \quad q_{\lambda\mu\nu} = 0 \quad (169)$$

Thus

$$\mathcal{L} = T + \frac{c_1}{\Lambda^2} T_\mu T^\mu \quad (170)$$

$$\text{Lowest-order non-vanishing operator: } T_\mu T^\mu \quad (171)$$

7 Local Lorentz Covariance and Absence of Preferred Frame

Covariant teleparallel structure:

$$T^a = de^a + \omega^a_b \wedge e^b, \quad R^a_b(\omega) = 0 \quad (172)$$

Local Lorentz transformations:

$$e^a_\mu \rightarrow e'^a_\mu = \Lambda^a_b(x) e^b_\mu \quad (173)$$

$$\omega^a_{b\mu} \rightarrow \omega'^a_{b\mu} = \Lambda^a_c \omega^c_{d\mu} (\Lambda^{-1})^d_b + \Lambda^a_c \partial_\mu (\Lambda^{-1})^c_b \quad (174)$$

Torsion covariance:

$$T'^a = de'^a + \omega'^a_b \wedge e'^b \quad (175)$$

$$T'^a = \Lambda^a_b T^b \quad (176)$$

Thus

$$T'^\rho_{\mu\nu} = T^\rho_{\mu\nu} \quad (177)$$

Scalar invariants:

$$T \rightarrow T, \quad T_\mu T^\mu \rightarrow T_\mu T^\mu \quad (178)$$

$$S[e, \omega] \text{ is invariant} \quad (179)$$

Drift parametrization:

$$e^i_0 = W^i, \quad W^i = b(t, x)x^i \quad (180)$$

Under local Lorentz transformation:

$$W^i \rightarrow W'^i = \Lambda^i_a e^a_0 \quad (181)$$

$$W'^i \neq W^i \quad (182)$$

Non-invariance of parametrization:

$$W^i \text{ is not a covariant object} \quad (183)$$

$$\text{Only } T^\rho_{\mu\nu}, T, T_\mu T^\mu \text{ are invariant} \quad (184)$$

No preferred frame condition:

Physical observables:

$$\mathcal{O} = \mathcal{O}(T^\rho_{\mu\nu}) \quad (185)$$

$$\mathcal{O}[e, \omega] = \mathcal{O}[e', \omega'] \quad (186)$$

Good tetrad consistency:

In inertial frames:

$$\omega^a_{b\mu} = 0 \quad (187)$$

In accelerated frames:

$$\omega^a_{b\mu} \neq 0 \quad (188)$$

$$T^a(e, \omega) = T^a(e', \omega') \quad (189)$$

Constraint nature of drift:

$$\frac{\partial \mathcal{L}}{\partial \dot{b}} = 0 \quad \Rightarrow \quad \pi_b = 0 \quad (190)$$

$$\frac{\partial \mathcal{L}}{\partial b} - \partial_i \left(\frac{\partial \mathcal{L}}{\partial (\partial_i b)} \right) = 0 \quad (191)$$

$$N_{\text{dof}}(b) = 0 \quad (192)$$

8 Variation of Action and the Field Equations

9 Variation of the Action

$$S = \frac{1}{2\kappa} \int d^4x e (T + \beta T_\mu T^\mu) \quad (193)$$

$$\delta S = \frac{1}{2\kappa} \int d^4x [\delta(eT) + \beta \delta(eT_\mu T^\mu)] \quad (194)$$

$$\delta e = e e_a{}^\mu \delta e^a{}_\mu \quad (195)$$

$$\delta(eT) = e \mathcal{E}_a{}^\nu \delta e^a{}_\nu \quad (196)$$

$$\delta(eT_\mu T^\mu) = \delta e T_\mu T^\mu + 2e T^\mu \delta T_\mu \quad (197)$$

$$\delta T_\mu = \delta(T^\nu{}_{\nu\mu}) \quad (198)$$

$$T^\rho{}_{\mu\nu} = e_a{}^\rho (\partial_\mu e^a{}_\nu - \partial_\nu e^a{}_\mu) \quad (199)$$

$$\delta T^\rho{}_{\mu\nu} = \delta e_a{}^\rho (\partial_\mu e^a{}_\nu - \partial_\nu e^a{}_\mu) + e_a{}^\rho (\partial_\mu \delta e^a{}_\nu - \partial_\nu \delta e^a{}_\mu) \quad (200)$$

$$\delta e_a{}^\rho = -e_a{}^\lambda e_b{}^\rho \delta e^b{}_\lambda \quad (201)$$

$$\delta T_\mu = \delta T^\nu{}_{\nu\mu} \quad (202)$$

$$\delta(eT_\mu T^\mu) = e (T_\alpha T^\alpha) e_a{}^\nu \delta e^a{}_\nu + 2e T^\mu \delta T_\mu \quad (203)$$

$$\delta S = \frac{1}{2\kappa} \int d^4x e [\mathcal{E}_a{}^\nu + \beta \mathcal{Z}_a{}^\nu] \delta e^a{}_\nu \quad (204)$$

$$\mathcal{E}_a{}^\nu + \beta \mathcal{Z}_a{}^\nu = \kappa \Theta_a{}^\nu \quad (205)$$

Variation of the $T_\mu T^\mu$ Sector

$$S_\beta = \frac{\beta}{2\kappa} \int d^4x e T_\mu T^\mu \quad (206)$$

$$\delta S_\beta = \frac{\beta}{2\kappa} \int d^4x [\delta e T_\alpha T^\alpha + 2e T^\mu \delta T_\mu + e T_\alpha T_\beta \delta g^{\alpha\beta}] \quad (207)$$

$$\delta e = e e_a{}^\nu \delta e^a{}_\nu \quad (208)$$

$$T_a := e_a{}^\mu T_\mu, \quad T^\lambda{}_{\mu a} := T^\lambda{}_{\mu\nu} e_a{}^\nu \quad (209)$$

$$\delta g_{\mu\nu} = \eta_{ab} (e^a{}_\mu \delta e^b{}_\nu + e^a{}_\nu \delta e^b{}_\mu) \quad (210)$$

$$\delta g^{\alpha\beta} = -g^{\alpha\mu} g^{\beta\nu} \delta g_{\mu\nu} \quad (211)$$

$$e T_\alpha T_\beta \delta g^{\alpha\beta} = -2e T^\nu T_a \delta e^a{}_\nu \quad (212)$$

$$T_\mu = T^\nu{}_{\nu\mu} \quad (213)$$

$$T^\rho{}_{\mu\nu} = e_a{}^\rho (\partial_\mu e^a{}_\nu - \partial_\nu e^a{}_\mu) \quad (214)$$

$$\delta T^\rho{}_{\mu\nu} = \delta e_a{}^\rho (\partial_\mu e^a{}_\nu - \partial_\nu e^a{}_\mu) + e_a{}^\rho (\partial_\mu \delta e^a{}_\nu - \partial_\nu \delta e^a{}_\mu) \quad (215)$$

$$\delta e_a{}^\rho = -e_a{}^\lambda e_b{}^\rho \delta e^b{}_\lambda \quad (216)$$

$$\delta T_\mu = \delta T^\nu{}_{\nu\mu} \quad (217)$$

$$\delta T_\mu = -T^\lambda{}_{\nu\mu} e_a{}^\nu \delta e^a{}_\lambda + \partial_\nu (e_a{}^\nu \delta e^a{}_\mu) - \partial_\mu (e_a{}^\nu \delta e^a{}_\nu) \quad (218)$$

$$\begin{aligned} 2e T^\mu \delta T_\mu &= -2e T^\mu T^\lambda{}_{\nu\mu} e_a{}^\nu \delta e^a{}_\lambda \\ &\quad + 2e T^\mu \partial_\nu (e_a{}^\nu \delta e^a{}_\mu) - 2e T^\mu \partial_\mu (e_a{}^\nu \delta e^a{}_\nu) \end{aligned} \quad (219)$$

$$2eT^\mu\partial_\nu(e_a{}^\nu\delta e^a{}_\mu)\rightarrow-2\partial_\nu(eT^\mu e_a{}^\nu)\delta e^a{}_\mu \quad (220)$$

$$-2eT^\mu\partial_\mu(e_a{}^\nu\delta e^a{}_\nu)\rightarrow+2\partial_\mu(eT^\mu e_a{}^\nu)\delta e^a{}_\nu \quad (221)$$

$$\delta S_\beta=\frac{\beta}{2\kappa}\int d^4x\,e\,Z_a{}^\nu\delta e^a{}_\nu \quad (222)$$

$$Z_a{}^\nu=e_a{}^\nu T_\alpha T^\alpha-2T^\nu T_a-2T^\mu T^\lambda{}_{\mu a}e_\lambda{}^\nu-2e^{-1}\partial_\mu(eT^\nu e_a{}^\mu)+2e^{-1}\partial_\mu(eT^\mu e_a{}^\nu) \quad (223)$$

$$e^{-1}\partial_\mu(eV^\mu)=\nabla_\mu V^\mu \quad (224)$$

$$Z_a{}^\nu=e_a{}^\nu T_\alpha T^\alpha-2T^\nu T_a-2T^\mu T^\lambda{}_{\mu a}e_\lambda{}^\nu-2\nabla_\mu(T^\nu e_a{}^\mu)+2\nabla_\mu(T^\mu e_a{}^\nu) \quad (225)$$

10 Solutions of Field Equations

10.1 Derivation of the Torsion Invariants

In this section, the explicit forms of the torsion invariants entering the action in the galactic limit are derived. In particular, the expressions $T_\mu T^\mu$ and T will be computed in terms of the drift function $W(r)$.

Under spherical symmetry, the drift field is defined as

$$W(r)=r\,b(r).$$

Furthermore, in the local galactic limit,

$$a=1, \quad \dot{a}=0, \quad c=1$$

will be assumed.

10.2 Derivation of $T_\mu T^\mu$

Our goal is to obtain the torsion-vector invariant $T_\mu T^\mu$ in terms of $W(r)$.

By definition, the torsion vector is

$$T_\mu=T^\nu{}_{\nu\mu}.$$

Direct calculation shows that only the temporal component is nonzero:

$$T_\mu = (T_0, 0, 0, 0).$$

As explicitly derived in the previous appendices, the resulting expression is

$$T_0 = \frac{r b_r(r) + 3(b - \dot{a})}{a},$$

where $b_r = \frac{db}{dr}$. Now let us write

$$b(r) = \frac{W(r)}{r}.$$

Then

$$b_r = \frac{d}{dr} \left(\frac{W}{r} \right) = \frac{W'r - W}{r^2}.$$

Therefore,

$$r b_r = W' - \frac{W}{r}.$$

Also,

$$3(b - \dot{a}) = \frac{3W}{r} - 3\dot{a}.$$

Substituting these expressions into T_0 gives

$$T_0 = \frac{W' - \frac{W}{r} + \frac{3W}{r} - 3\dot{a}}{a},$$

and after simplification,

$$T_0 = \frac{W' + \frac{2W}{r} - 3\dot{a}}{a}.$$

We now compute the invariant:

$$T^\mu T_\mu = g^{\mu\nu} T_\mu T_\nu.$$

Since only $T_0 \neq 0$,

$$T^\mu T_\mu = g^{00} T_0^2.$$

From the inverse metric,

$$g^{00} = -\frac{1}{c^2},$$

and therefore

$$T^\mu T_\mu = -\frac{T_0^2}{c^2}.$$

Hence,

$$T^\mu T_\mu = -\frac{\left(W' + \frac{2W}{r} - 3\dot{a}\right)^2}{a^2 c^2}.$$

In the local galactic limit,

$$a = 1, \quad \dot{a} = 0, \quad c = 1,$$

the result becomes

$$\boxed{T^\mu T_\mu = -\left(W' + \frac{2W}{r}\right)^2}.$$

10.3 Derivation of the Torsion Scalar T

We now compute the TEGR torsion scalar T in terms of $W(r)$.

As explicitly derived in previous appendices, the general expression is

$$T = \frac{2(b - \dot{a})(2rb_r + 3b - 3\dot{a})}{a^2 c^2}.$$

Again, let

$$b(r) = \frac{W(r)}{r}.$$

First,

$$b_r = \frac{W'r - W}{r^2},$$

which implies

$$2rb_r = 2W' - \frac{2W}{r}.$$

Also,

$$3b = \frac{3W}{r}.$$

Therefore, the expression inside the parentheses becomes

$$2rb_r + 3b = 2W' - \frac{2W}{r} + \frac{3W}{r},$$

which simplifies to

$$2rb_r + 3b = 2W' + \frac{W}{r}.$$

Thus the torsion scalar becomes

$$T = \frac{2\frac{W}{r} \left(2W' + \frac{W}{r}\right)}{a^2 c^2}.$$

In the local galactic limit,

$$a = 1, \quad c = 1,$$

the final expression is

$$\boxed{T = 2\frac{W(r)}{r} \left(2W'(r) + \frac{W(r)}{r}\right)}.$$

10.4 Derivation of Effective Radial Lagrangian

In this case, the Lagrangian density of the action is

$$\mathcal{L} = T + \beta T^\mu T_\mu$$

Using the results obtained in the previous sections,

$$T = 2\frac{W}{r} \left(2W' + \frac{W}{r}\right)$$

and

$$T^\mu T_\mu = - \left(W' + \frac{2W}{r}\right)^2,$$

we obtain

$$\mathcal{L} = 2\frac{W}{r} \left(2W' + \frac{W}{r}\right) - \beta \left(W' + \frac{2W}{r}\right)^2.$$

Expanding the first term gives

$$2\frac{W}{r} \left(2W' + \frac{W}{r}\right) = \frac{4WW'}{r} + \frac{2W^2}{r^2}.$$

For the second term,

$$\left(W' + \frac{2W}{r}\right)^2 = W'^2 + \frac{4WW'}{r} + \frac{4W^2}{r^2}.$$

Therefore, the Lagrangian density becomes

$$\mathcal{L} = \frac{4WW'}{r} + \frac{2W^2}{r^2} - \beta \left(W'^2 + \frac{4WW'}{r} + \frac{4W^2}{r^2} \right).$$

Combining similar terms yields

$$\boxed{\mathcal{L} = -\beta W'^2 + 4(1 - \beta) \frac{WW'}{r} + (2 - 4\beta) \frac{W^2}{r^2}}.$$

Under spherical symmetry, the volume element is

$$d^3x = 4\pi r^2 dr,$$

so the effective radial Lagrangian can be written as

$$L = r^2 \mathcal{L}.$$

Consequently,

$$\boxed{L = -\beta r^2 W'^2 + 4(1 - \beta) r W W' + (2 - 4\beta) W^2}$$

is obtained.

10.5 Euler-Lagrange Solution of $W(r)$

$$L = -\beta r^2 W'^2 + 4(1 - \beta) r W W' + (2 - 4\beta) W^2.$$

The Euler-Lagrange equation is

$$\frac{d}{dr} \left(\frac{\partial L}{\partial W'} \right) - \frac{\partial L}{\partial W} = 0.$$

First, let us take the derivative with respect to W' :

$$\frac{\partial L}{\partial W'} = -2\beta r^2 W' + 4(1 - \beta) r W.$$

Now differentiate this expression with respect to r :

$$\frac{d}{dr} \left(\frac{\partial L}{\partial W'} \right) = \frac{d}{dr} [-2\beta r^2 W' + 4(1 - \beta) r W].$$

This gives

$$\frac{d}{dr} \left(\frac{\partial L}{\partial W'} \right) = -2\beta (2rW' + r^2 W'') + 4(1 - \beta) (W + rW').$$

On the other hand, the derivative with respect to W is

$$\frac{\partial L}{\partial W} = 4(1 - \beta)rW' + 2(2 - 4\beta)W.$$

That is,

$$\frac{\partial L}{\partial W} = 4(1 - \beta)rW' + (4 - 8\beta)W.$$

Now substitute these expressions into the Euler–Lagrange equation:

$$-2\beta (2rW' + r^2W'') + 4(1 - \beta) (W + rW') - 4(1 - \beta)rW' - (4 - 8\beta)W = 0.$$

Expanding,

$$-4\beta rW' - 2\beta r^2W'' + 4(1 - \beta)W + 4(1 - \beta)rW' - 4(1 - \beta)rW' - (4 - 8\beta)W = 0.$$

Here,

$$4(1 - \beta)rW' - 4(1 - \beta)rW' = 0$$

cancels out. The remaining equation is

$$-2\beta r^2W'' - 4\beta rW' + 4\beta W = 0.$$

Factoring out the common factor -2β ,

$$-2\beta (r^2W'' + 2rW' - 2W) = 0.$$

Therefore, for $\beta \neq 0$, the resulting equation is

$$\boxed{r^2W'' + 2rW' - 2W = 0}.$$

10.6 Vacuum Solution for $W(r)$

In the vacuum solution, since matter is absent from the field equations,

$$\rho = 0$$

is imposed. In this case, the Euler–Lagrange equation becomes

$$\boxed{r^2W'' + 2rW' - 2W = 0}.$$

This differential equation can be solved using a power-law ansatz. Let us propose a solution of the form

$$W(r) = r^m.$$

Then the derivatives become

$$W' = mr^{m-1}$$

and

$$W'' = m(m-1)r^{m-2}.$$

Substituting these into the differential equation gives

$$r^2m(m-1)r^{m-2} + 2rmr^{m-1} - 2r^m = 0.$$

Simplifying the powers yields

$$m(m-1)r^m + 2mr^m - 2r^m = 0.$$

Factoring out r^m ,

$$r^m (m^2 + m - 2) = 0.$$

Since the trivial solution

$$r^m = 0$$

is not physical, the characteristic equation becomes

$$m^2 + m - 2 = 0.$$

Factoring this equation gives

$$(m+2)(m-1) = 0.$$

Therefore, the roots are

$$m = -2, \quad m = 1.$$

Thus the general vacuum solution is obtained as

$$\boxed{W(r) = C_1 r + \frac{C_2}{r^2}}.$$

10.7 Vacuum Constraint Equation

However, obtaining the vacuum solution from the Euler–Lagrange equation alone is not sufficient. Additional constraints arising from the field equations must also be satisfied. For this purpose, the $(0, 0)$ component of the field equations must be examined.

The general expression is

$$\frac{\beta (rb_r + 3(b - \dot{a}))^2 + 2(\dot{a} - b)(2rb_r - 3(\dot{a} - b))}{4a^2} = \frac{\kappa\rho}{2}.$$

Since we seek a vacuum solution,

$$\rho = 0$$

is imposed, and the equation becomes

$$\beta (rb_r + 3(b - \dot{a}))^2 + 2(\dot{a} - b)(2rb_r - 3(\dot{a} - b)) = 0.$$

In the local galactic limit,

$$a = 1, \quad \dot{a} = 0,$$

this reduces to

$$\beta (rb_r + 3b)^2 - 2b(2rb_r + 3b) = 0.$$

Now let us use the definition

$$W(r) = rb(r).$$

From this,

$$b(r) = \frac{W(r)}{r}$$

and

$$b_r = \frac{d}{dr} \left(\frac{W}{r} \right) = \frac{W'r - W}{r^2}$$

are obtained.

Therefore,

$$rb_r = W' - \frac{W}{r}.$$

Thus,

$$rb_r + 3b = W' - \frac{W}{r} + \frac{3W}{r} = W' + \frac{2W}{r},$$

and similarly,

$$2rb_r + 3b = 2W' - \frac{2W}{r} + \frac{3W}{r} = 2W' + \frac{W}{r}.$$

Substituting these into the field equation gives

$$\beta \left(W' + \frac{2W}{r} \right)^2 - 2 \frac{W}{r} \left(2W' + \frac{W}{r} \right) = 0.$$

Consequently, the constraint equation that the vacuum solution must satisfy is

$$\boxed{\beta \left(W' + \frac{2W}{r} \right)^2 = 2 \frac{W}{r} \left(2W' + \frac{W}{r} \right)}.$$

10.8 Consistency of the Vacuum Solution

The vacuum solution obtained in the previous section was

$$W(r) = C_1 r + \frac{C_2}{r^2}.$$

This solution must also satisfy the vacuum constraint arising from the field equations. For this purpose, the solution is substituted into

$$\beta \left(W' + \frac{2W}{r} \right)^2 = 2 \frac{W}{r} \left(2W' + \frac{W}{r} \right)$$

First, the derivative is computed:

$$W' = C_1 - \frac{2C_2}{r^3}.$$

Also,

$$\frac{W}{r} = C_1 + \frac{C_2}{r^3}$$

is obtained.

Therefore,

$$W' + \frac{2W}{r} = \left(C_1 - \frac{2C_2}{r^3} \right) + 2 \left(C_1 + \frac{C_2}{r^3} \right)$$

and after simplification,

$$W' + \frac{2W}{r} = 3C_1.$$

Similarly,

$$2W' + \frac{W}{r} = 2 \left(C_1 - \frac{2C_2}{r^3} \right) + \left(C_1 + \frac{C_2}{r^3} \right)$$

which gives

$$2W' + \frac{W}{r} = 3C_1 - \frac{3C_2}{r^3}.$$

Substituting these expressions into the constraint equation yields

$$\beta(3C_1)^2 = 2 \left(C_1 + \frac{C_2}{r^3} \right) \left(3C_1 - \frac{3C_2}{r^3} \right).$$

Expanding the right-hand side gives

$$9\beta C_1^2 = 6 \left(C_1^2 - \frac{C_2^2}{r^6} \right).$$

For this expression to hold for all r , the term

$$\frac{C_2^2}{r^6}$$

must vanish. Therefore,

$$\boxed{C_2 = 0}$$

is obtained.

In this case, the equation reduces to

$$9\beta C_1^2 = 6C_1^2,$$

which implies

$$(9\beta - 6)C_1^2 = 0.$$

Thus two possibilities emerge:

$$\boxed{\beta = \frac{2}{3}}$$

or

$$\boxed{C_1 = 0.}$$

In the second case, since additionally

$$C_2 = 0,$$

one obtains

$$\boxed{W(r) = 0}.$$

Therefore, for a nontrivial vacuum solution to exist, the theory parameter must satisfy

$$\beta = \frac{2}{3}.$$

However, this value is forbidden due to the consistency constraints derived in the previous sections. Consequently,

$$C_1 = 0, \quad C_2 = 0, \quad W(r) = 0$$

is obtained.

10.9 Matter Sector Solution

In this section, the solution in the matter sector is investigated. For this purpose, the $(0, 0)$ and $(1, 0)$ components of the field equations will be used.

The $(0, 0)$ Equation

The $(0, 0)$ component of the field equations is

$$\frac{\beta (rb_r + 3b)^2 - 2b(2rb_r + 3b)}{4} = \frac{\kappa\rho}{2} (r^2b^2 + rWb - 1).$$

Using the definition $W = rb$,

$$r^2b^2 + rWb - 1 = r^2b^2 + r^2b^2 - 1 = 2r^2b^2 - 1 = 2W^2 - 1$$

is obtained.

Furthermore, as shown previously,

$$rb_r + 3b = W' + \frac{2W}{r}$$

and

$$2rb_r + 3b = 2W' + \frac{W}{r}.$$

Therefore, the equation becomes

$$\frac{\beta \left(W' + \frac{2W}{r}\right)^2 - 2\frac{W}{r} \left(2W' + \frac{W}{r}\right)}{4} = \frac{\kappa\rho}{2} (2W^2 - 1).$$

From this, the matter density is obtained as

$$\rho(r) = \frac{\beta \left(W' + \frac{2W}{r}\right)^2 - 2\frac{W}{r} \left(2W' + \frac{W}{r}\right)}{2\kappa(2W^2 - 1)}.$$

The (1, 0) Equation

The (1, 0) component of the field equations is

$$\frac{\beta}{2r^3} \partial_r (r^4 b_r) = -\frac{\kappa\rho}{2} [r^3 b^3 - r^2 W b^2 - r b - W^2].$$

Now let us use the definition

$$W = rb.$$

First,

$$r^2 W b^2 = W (rb)^2 = W^3$$

and

$$r^3 b^3 = W^3$$

show that the first two terms cancel each other.

Also,

$$rb = W,$$

so the right-hand side becomes

$$-\frac{\kappa\rho}{2} (-W - W^2),$$

and consequently,

$$\frac{\beta}{2r^3} \partial_r (r^4 b_r) = \frac{\kappa\rho}{2} W(1 + W).$$

Now let us compute the derivative term.

First write

$$b(r) = \frac{W(r)}{r}.$$

Therefore,

$$b_r = \frac{d}{dr} \left(\frac{W}{r} \right) = \frac{W'r - W}{r^2}.$$

From this,

$$r^4 b_r = r^2(W'r - W) = r^3 W' - r^2 W$$

is obtained.

Taking the derivative gives

$$\begin{aligned}\partial_r(r^4 b_r) &= \partial_r(r^3 W' - r^2 W) \\ &= 3r^2 W' + r^3 W'' - 2rW - r^2 W' \\ &= r^3 W'' + 2r^2 W' - 2rW.\end{aligned}$$

Therefore,

$$\frac{\beta}{2r^3} \partial_r(r^4 b_r) = \frac{\beta}{2} \left(W'' + \frac{2W'}{r} - \frac{2W}{r^2} \right).$$

Consequently, the (1, 0) equation becomes

$$\boxed{\beta \left(W'' + \frac{2W'}{r} - \frac{2W}{r^2} \right) = \kappa \rho W(1 + W)}.$$

Combined Matter Equation

Finally, substituting the expression obtained from the (0, 0) equation,

$$\rho(r) = \frac{\beta \left(W' + \frac{2W}{r} \right)^2 - 2\frac{W}{r} \left(2W' + \frac{W}{r} \right)}{2\kappa(2W^2 - 1)},$$

into the (1, 0) equation gives

$$\beta \left(W'' + \frac{2W'}{r} - \frac{2W}{r^2} \right) = \kappa W(1 + W) \left(\frac{\beta \left(W' + \frac{2W}{r} \right)^2 - 2\frac{W}{r} \left(2W' + \frac{W}{r} \right)}{2\kappa(2W^2 - 1)} \right).$$

Thus, the drift equation in the matter sector is obtained as

$$\boxed{\beta \left(W'' + \frac{2W'}{r} - \frac{2W}{r^2} \right) = W(1 + W) \left(\frac{\beta \left(W' + \frac{2W}{r} \right)^2 - 2\frac{W}{r} \left(2W' + \frac{W}{r} \right)}{2(2W^2 - 1)} \right)}.$$

10.10 Power Series Solution Near the Galactic Center

In this section, the regular solution near the galactic center for the differential equation obtained in the matter sector is investigated.

The equation to be solved is

$$\beta \left(W'' + \frac{2W'}{r} - \frac{2W}{r^2} \right) = \frac{W(1+W)}{2(2W^2-1)} \left[\beta \left(W' + \frac{2W}{r} \right)^2 - 2\frac{W}{r} \left(2W' + \frac{W}{r} \right) \right].$$

To obtain a regular solution at the center, the most natural choice is

$$W(0) = 0,$$

because the equation contains terms such as

$$\frac{W}{r}, \quad \frac{W'}{r}.$$

For this reason, the solution is written as a power series:

$$W(r) = a_1 r + a_2 r^2 + a_3 r^3 + a_4 r^4 + \dots$$

Substituting the series into the differential equation and matching coefficients first yields

$$a_2 = 0.$$

The higher-order coefficients are found to be

$$a_3 = \frac{3a_1^2(2-3\beta)}{20\beta}$$

$$a_4 = \frac{a_1^4(2-3\beta)}{12\beta}$$

and

$$a_5 = \frac{3a_1^5(-3\beta^2 - 76\beta + 52)}{1120\beta^2}.$$

Therefore, the solution near the center takes the form

$$W(r) = a_1 r + \frac{3a_1^2(2-3\beta)}{20\beta} r^3 + \frac{a_1^4(2-3\beta)}{12\beta} r^4 + \frac{3a_1^5(-3\beta^2 - 76\beta + 52)}{1120\beta^2} r^5 + \dots$$

where a_1 is the free initial slope:

$$\boxed{a_1 = W'(0)}.$$

Thus, at leading order near the center, the solution behaves as

$$\boxed{W(r) \sim a_1 r}.$$

This result is highly important. Since

$$b(r) = \frac{W(r)}{r},$$

one obtains near the center

$$b(r) \sim a_1,$$

and therefore $b(r)$ remains finite at the center.

Similarly, the density also remains finite at the center. At leading order,

$$\rho(0) = -\frac{a_1^2(9\beta - 6)}{2\kappa}$$

or equivalently,

$$\boxed{\rho(0) = \frac{3a_1^2(2 - 3\beta)}{2\kappa}}.$$

Physically, a positive central density requires the condition

$$\boxed{\beta < \frac{2}{3}}.$$

11 Full Set of Field Equations

The field equations are obtained by varying the action with respect to the tetrad field $e^a{}_\mu$. The resulting equations take the form

$$\mathcal{E}_a{}^\nu + \frac{\beta}{4}\mathcal{Z}_a{}^\nu = \frac{\kappa}{2}\Theta_a{}^\nu. \quad (226)$$

Here, $\mathcal{E}_a{}^\nu$ denotes the standard TEGR contribution, $\mathcal{Z}_a{}^\nu$ represents the modification generated by the $T_\mu T^\mu$ sector, and $\Theta_a{}^\nu$ is the mixed tetrad-coordinate matter tensor entering the field equations.

For the drift-fluid configuration considered in this work, the matter sector is defined in the coordinate basis as

$$\Theta_\mu{}^\nu = \text{diag}(-\rho, p, p, p). \quad (227)$$

Projecting with the inverse tetrad,

$$\Theta_a{}^\nu = e_a{}^\mu \Theta_\mu{}^\nu,$$

one obtains the mixed components entering the field equations. For the radial drift tetrad

$$W(r) = rb(r),$$

the nonvanishing components become

$$\Theta_a{}^\nu = \begin{pmatrix} \rho & 0 & 0 & 0 \\ \frac{rb(r)\rho}{a(t)} & \frac{p}{a(t)} & 0 & 0 \\ 0 & 0 & \frac{p}{ra(t)} & 0 \\ 0 & 0 & 0 & \frac{p}{ra(t)\sin\theta} \end{pmatrix}. \quad (228)$$

Therefore, the independent field equations may be written as follows.

• Temporal Component

$$(0,0): \quad \mathcal{E}_0{}^0 + \beta \mathcal{Z}_0{}^0 = \frac{\kappa}{2} \rho \quad (229)$$

Drift Energy-Flux Equation

$$(1,0): \quad \mathcal{E}_1{}^0 + \beta \mathcal{Z}_1{}^0 = \frac{\kappa}{2} \frac{rb(r)\rho}{a(t)} \quad (230)$$

This equation determines the coupling between the drift field and the matter-energy density.

Radial Pressure Equation

$$(1,1): \quad \mathcal{E}_1{}^1 + \beta \mathcal{Z}_1{}^1 = \frac{\kappa}{2} \frac{p}{a(t)} \quad (231)$$

Angular Pressure Equations

$$(2,2): \quad \mathcal{E}_2{}^2 + \beta \mathcal{Z}_2{}^2 = \frac{\kappa}{2} \frac{p}{ra(t)} \quad (232)$$

$$(3,3): \quad \mathcal{E}_3{}^3 + \beta \mathcal{Z}_3{}^3 = \frac{\kappa}{2} \frac{p}{ra(t)\sin\theta} \quad (233)$$

The remaining off-diagonal equations vanish identically due to spherical symmetry and the absence of anisotropic stresses in the matter sector.

11.1 Explicit Versions of Field Equations

For the radial drift tetrad

$$W(r) = rb(t, r),$$

the field equations are

$$\mathcal{E}_a{}^\nu + \beta \mathcal{Z}_a{}^\nu = \frac{\kappa}{2} \Theta_a{}^\nu. \quad (234)$$

The nonvanishing matter components are

$$\Theta_a{}^\nu = \begin{pmatrix} -\rho & 0 & 0 & 0 \\ \frac{rb\rho}{a} & \frac{p}{a} & 0 & 0 \\ 0 & 0 & \frac{p}{ra} & 0 \\ 0 & 0 & 0 & \frac{p}{ra \sin \theta} \end{pmatrix}. \quad (235)$$

11.1.1 The (0, 0) Equation

$$\begin{aligned} & \frac{\beta r^2 b_r^2 + 6\beta r b b_r - 6\beta r \dot{a} b_r + 9\beta b^2 - 18\beta b \dot{a} + 9\beta \dot{a}^2}{4a^2} \\ & - \frac{12rb b_r - 12r \dot{a} b_r + 18b^2 - 36b \dot{a} + 18\dot{a}^2}{4a^2} = -\frac{\kappa}{2} \rho. \end{aligned} \quad (236)$$

11.1.2 The (0, 1) Equation

$$\begin{aligned} & \frac{r b}{4a^3} \left[-\beta r^2 b_r^2 - 6\beta r b b_r + 6\beta r \dot{a} b_r - 9\beta b^2 + 18\beta b \dot{a} - 9\beta \dot{a}^2 \right. \\ & \left. + 12rb b_r - 12r \dot{a} b_r + 18b^2 - 36b \dot{a} + 18\dot{a}^2 \right] = 0. \end{aligned} \quad (237)$$

11.1.3 The (1, 0) Equation

$$\boxed{\frac{\beta}{2a^2} (rb_{rr} + 4b_r) = \frac{-\kappa rb\rho}{2a}} \quad (238)$$

or equivalently,

$$\boxed{\beta (rb_{rr} + 4b_r) \kappa a r b \rho = 0.} \quad (239)$$

11.1.4 The (1, 1) Equation

$$\mathcal{E}_1^1 + \beta \mathcal{Z}_1^1 = \frac{\kappa}{2a} p. \quad (240)$$

11.1.5 The (2, 2) Equation

$$\mathcal{E}_2^2 + \beta \mathcal{Z}_2^2 = \frac{\kappa}{2ra} p. \quad (241)$$

11.1.6 The (3, 3) Equation

$$\mathcal{E}_3^3 + \beta \mathcal{Z}_3^3 = \frac{\kappa}{2ra \sin \theta} p. \quad (242)$$

The angular consistency condition is

$$\mathcal{E}_3^3 \sin \theta - \mathcal{E}_2^2 = 0, \quad (243)$$

showing that the angular equations are not independent.

12 Solutions of Modified Equations and Fundamental Variables

12.1 Derivation of ρ_{drift}

$$3 \left(1 - \frac{3}{2} \beta \right) \mathcal{H}_{\text{eff}}^2 - \frac{1}{3} Y^2 = \kappa \rho \quad (244)$$

$$3H^2 = \kappa(\rho + \rho_{\text{drift}}) \quad (245)$$

$$\kappa \rho_{\text{drift}} = 3H^2 - \kappa \rho \quad (246)$$

$$\kappa \rho = 3 \left(1 - \frac{3}{2} \beta \right) \mathcal{H}_{\text{eff}}^2 - \frac{1}{3} Y^2 \quad (247)$$

$$\kappa \rho_{\text{drift}} = 3H^2 - \left[3 \left(1 - \frac{3}{2} \beta \right) \mathcal{H}_{\text{eff}}^2 - \frac{1}{3} Y^2 \right] \quad (248)$$

$$= 3H^2 - 3 \left(1 - \frac{3}{2} \beta \right) \mathcal{H}_{\text{eff}}^2 + \frac{1}{3} Y^2 \quad (249)$$

$$\rho_{\text{drift}} = \frac{3}{\kappa} \left[H^2 - \left(1 - \frac{3}{2} \beta \right) \mathcal{H}_{\text{eff}}^2 + \frac{1}{9} Y^2 \right] \quad (250)$$

$$\mathcal{H}_{\text{eff}} = H - \frac{b}{a} - \frac{1}{3}Y \quad (251)$$

$$\mathcal{H}_{\text{eff}}^2 = \left(H - \frac{b}{a} - \frac{1}{3}Y \right)^2 \quad (252)$$

$$= H^2 + \frac{b^2}{a^2} + \frac{1}{9}Y^2 - 2H\frac{b}{a} - \frac{2}{3}HY + \frac{2}{3}\frac{b}{a}Y \quad (253)$$

$$\rho_{\text{drift}} = \frac{3}{\kappa} \left[H^2 - \left(1 - \frac{3}{2}\beta \right) \left(H^2 + \frac{b^2}{a^2} + \frac{1}{9}Y^2 - 2H\frac{b}{a} - \frac{2}{3}HY + \frac{2}{3}\frac{b}{a}Y \right) + \frac{1}{9}Y^2 \right] \quad (254)$$

12.2 Derivation of p_{drift}

$$\left(1 - \frac{3}{2}\beta \right) \left(2\dot{\mathcal{H}}_{\text{eff}} + 3\mathcal{H}_{\text{eff}}^2 \right) + \frac{1}{3}Y^2 = -\kappa p \quad (255)$$

$$2\dot{H} + 3H^2 = -\kappa(p + p_{\text{drift}}) \quad (256)$$

$$-\kappa p_{\text{drift}} = 2\dot{H} + 3H^2 + \kappa p \quad (257)$$

$$\kappa p = - \left(1 - \frac{3}{2}\beta \right) \left(2\dot{\mathcal{H}}_{\text{eff}} + 3\mathcal{H}_{\text{eff}}^2 \right) - \frac{1}{3}Y^2 \quad (258)$$

$$-\kappa p_{\text{drift}} = 2\dot{H} + 3H^2 - \left(1 - \frac{3}{2}\beta \right) \left(2\dot{\mathcal{H}}_{\text{eff}} + 3\mathcal{H}_{\text{eff}}^2 \right) - \frac{1}{3}Y^2 \quad (259)$$

$$p_{\text{drift}} = \frac{1}{\kappa} \left[\left(1 - \frac{3}{2}\beta \right) \left(2\dot{\mathcal{H}}_{\text{eff}} + 3\mathcal{H}_{\text{eff}}^2 \right) + \frac{1}{3}Y^2 - (2\dot{H} + 3H^2) \right] \quad (260)$$

For homogeneous regime,

$$\partial_i b = 0 \quad (261)$$

$$Y = 0 \quad (262)$$

$$\mathcal{H}_{\text{eff}} = H - \frac{b}{a} \quad (263)$$

$$\rho_{\text{drift}} = \frac{3}{\kappa} \left[H^2 - \left(1 - \frac{3}{2}\beta \right) H_{\text{eff}}^2 \right] \quad (264)$$

$$p_{\text{drift}} = \frac{1}{\kappa} \left[\left(1 - \frac{3}{2}\beta \right) (2\dot{H}_{\text{eff}} + 3H_{\text{eff}}^2) - (2\dot{H} + 3H^2) \right] \quad (265)$$

12.3 Derivation of $\dot{\mathcal{H}}_{\text{eff}}$

$$\mathcal{H}_{\text{eff}} = H - \frac{b}{a} - \frac{1}{3}Y \quad (266)$$

$$\dot{\mathcal{H}}_{\text{eff}} = \dot{H} - \frac{d}{dt} \left(\frac{b}{a} \right) - \frac{1}{3}\dot{Y} \quad (267)$$

$$\frac{d}{dt} \left(\frac{b}{a} \right) = \frac{\dot{b}}{a} - H \frac{b}{a} \quad (268)$$

$$\dot{Y} = \frac{x^i \partial_i \dot{b}}{a} - HY \quad (269)$$

$$\dot{\mathcal{H}}_{\text{eff}} = \dot{H} - \frac{\dot{b}}{a} + H \frac{b}{a} - \frac{1}{3} \left(\frac{x^i \partial_i \dot{b}}{a} - HY \right) \quad (270)$$

13 Derivation of Limits

13.1 Newtonian Limit

In this section, we derive the Newtonian limit of the drift framework and show how the standard Newtonian gravitational potential emerges from the drift field.

We begin with the tetrad

$$e^0 = dt, \quad e^i = dx^i + W^i dt.$$

In the weak-field and nonrelativistic regime, cosmological effects are neglected, so that

$$a \rightarrow 1, \quad b \rightarrow 0, \quad \partial_t W^i \rightarrow 0.$$

We define the effective physical velocity field as

$$v^i := u^i + W^i,$$

where u^i is the ordinary particle velocity and W^i is the drift flow.

For a particle comoving with the drift frame,

$$v^i = 0,$$

which implies

$$u^i = -W^i.$$

Assuming inertial motion in the drift frame,

$$\frac{dv^i}{dt} = 0.$$

Substituting the definition of v^i ,

$$\frac{d}{dt} (u^i + W^i) = 0,$$

gives

$$\frac{du^i}{dt} + \frac{dW^i}{dt} = 0.$$

Using the total derivative,

$$\frac{dW^i}{dt} = \frac{\partial W^i}{\partial t} + \frac{dx^j}{dt} \frac{\partial W^i}{\partial x^j},$$

we obtain

$$\frac{dW^i}{dt} = \partial_t W^i + u^j \partial_j W^i.$$

Since the drift field is static in the Newtonian regime,

$$\partial_t W^i = 0,$$

it follows that

$$\frac{dW^i}{dt} = u^j \partial_j W^i.$$

Using

$$u^j = -W^j,$$

we find

$$\frac{dW^i}{dt} = -W^j \partial_j W^i.$$

Therefore,

$$\frac{du^i}{dt} - W^j \partial_j W^i = 0,$$

or equivalently,

$$\boxed{\frac{du^i}{dt} = W^j \partial_j W^i}$$

This equation plays the role of the effective gravitational acceleration. In Newtonian gravity,

$$\frac{du^i}{dt} = -\partial_i \Phi,$$

so comparison gives

$$-\partial_i \Phi = W^j \partial_j W^i.$$

Using the vector identity

$$(\mathbf{W} \cdot \nabla) \mathbf{W} = \nabla \left(\frac{W^2}{2} \right) - \mathbf{W} \times (\nabla \times \mathbf{W}),$$

together with the irrotational condition

$$\nabla \times \mathbf{W} = 0,$$

we obtain

$$(\mathbf{W} \cdot \nabla) \mathbf{W} = \nabla \left(\frac{W^2}{2} \right).$$

Hence,

$$W^j \partial_j W^i = \partial_i \left(\frac{W^2}{2} \right).$$

Therefore,

$$-\partial_i \Phi = \partial_i \left(\frac{W^2}{2} \right),$$

which implies

$$\partial_i \left(\Phi + \frac{W^2}{2} \right) = 0.$$

Thus,

$$\Phi + \frac{W^2}{2} = \text{const.}$$

Imposing asymptotic flatness,

$$\Phi(\infty) = 0, \quad W(\infty) = 0,$$

gives

$$\text{const} = 0.$$

Therefore,

$$\boxed{\Phi = -\frac{W^2}{2}}$$

is obtained.

This result is central to the drift interpretation of gravity: the Newtonian potential emerges directly from the norm of the drift field.

The gravitational acceleration is then

$$\mathbf{g} = -\nabla\Phi,$$

or equivalently,

$$\mathbf{g} = \nabla \left(\frac{W^2}{2} \right).$$

To connect with the Poisson equation, consider

$$\nabla^2\Phi = 4\pi G\rho.$$

Substituting the drift potential gives

$$\nabla^2 \left(-\frac{W^2}{2} \right) = 4\pi G\rho,$$

which implies

$$\nabla^2 W^2 = -8\pi G\rho.$$

In vacuum,

$$\rho = 0,$$

and therefore

$$\nabla^2 W^2 = 0.$$

For a spherically symmetric drift field,

$$W^i = W(r)n^i, \quad n^i = \frac{x^i}{r},$$

one has

$$W^2 = W(r)^2.$$

The Laplacian becomes

$$\nabla^2 W^2 = \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dW^2}{dr} \right).$$

Thus,

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{dW^2}{dr} \right) = 0,$$

which integrates to

$$r^2 \frac{dW^2}{dr} = C_1.$$

Hence,

$$\frac{dW^2}{dr} = \frac{C_1}{r^2},$$

and integrating once more gives

$$W^2 = C_2 - \frac{C_1}{r}.$$

Using the boundary condition

$$W(\infty) = 0 \quad \Rightarrow \quad C_2 = 0,$$

we obtain

$$W^2 = -\frac{C_1}{r}.$$

Identifying the integration constant with the physical mass parameter yields

$$W^2 = \frac{2GM}{r}.$$

Consequently,

$$\Phi = -\frac{W^2}{2} = -\frac{GM}{r}.$$

This is exactly the Newtonian gravitational potential.

Finally, the resulting acceleration becomes

$$g^r = -\frac{GM}{r^2},$$

so that

$$\boxed{\frac{du^r}{dt} = -\frac{GM}{r^2}}$$

is recovered exactly.

Therefore, the drift framework reproduces the standard Newtonian limit in vacuum, with the gravitational potential emerging dynamically from the radial drift flow.

13.2 Derivation of the Poisson-Type Structure

In this section, we derive the effective Poisson-type structure generated by the drift field and show how a Newtonian-like gravitational potential naturally emerges from the torsional drift sector.

We begin by defining the effective scalar potential associated with the drift field:

$$\Phi = \frac{1}{2}W^iW_i.$$

Taking a spatial derivative gives

$$\partial_i\Phi = \frac{1}{2}\partial_i(W^jW_j).$$

Using the product rule,

$$\partial_i\Phi = \frac{1}{2}\left((\partial_iW^j)W_j + W^j\partial_iW_j\right).$$

Since the two terms are identical after index contraction, this simplifies to

$$\partial_i\Phi = W^j\partial_iW_j.$$

For irrotational drift configurations, the derivative indices may be exchanged, yielding

$$\partial_i\Phi = W^j\partial_jW_i.$$

We now compute the Laplacian of the potential:

$$\nabla^2\Phi = \partial_i\partial_i\Phi.$$

Substituting the previous relation gives

$$\nabla^2\Phi = \partial_i(W^j\partial_jW_i).$$

Applying the product rule again yields

$$\nabla^2\Phi = (\partial_i W^j)(\partial_j W_i) + W^j \partial_i \partial_j W_i.$$

Since ordinary derivatives commute,

$$\nabla^2\Phi = (\partial_i W^j)(\partial_j W_i) + W^j \partial_j (\partial_i W_i).$$

At this stage, we define the Newtonian-limit torsion trace

$$T_0^{(N)} := \partial_i W^i.$$

Therefore,

$$\nabla^2\Phi = (\partial_i W^j)(\partial_j W_i) + W^j \partial_j T_0^{(N)}.$$

Hence the effective Poisson-type relation becomes

$$\boxed{\nabla^2\Phi = (\partial_i W^j)(\partial_j W_i) + W^j \partial_j T_0^{(N)}}$$

This equation is highly important because it shows that the gravitational potential is sourced directly by geometric drift quantities constructed from the torsion sector.

We now consider the simplest homogeneous drift configuration:

$$W^i = \alpha x^i.$$

Its derivative is

$$\partial_j W^i = \alpha \delta_j^i.$$

The torsion trace becomes

$$T_0^{(N)} = \partial_i W^i = 3\alpha.$$

Since α is constant,

$$\partial_j T_0^{(N)} = 0.$$

The quadratic derivative term evaluates to

$$(\partial_i W^j)(\partial_j W_i) = \alpha^2 \delta_i^j \delta_j^i = 3\alpha^2.$$

Therefore,

$$\nabla^2\Phi = 3\alpha^2.$$

Thus the Laplacian of the potential becomes constant:

$$\nabla^2\Phi = \text{const.}$$

Comparing with the standard Newtonian Poisson equation,

$$\nabla^2\Phi = 4\pi G\rho,$$

one obtains an effective density

$$\rho = \frac{3\alpha^2}{4\pi G}.$$

Consequently,

$$\boxed{\nabla^2\Phi = 4\pi G\rho}$$

is recovered directly from the drift geometry.

This result is remarkable because it shows that a Newtonian-like gravitational source can emerge purely from the structure of the torsional drift field, without introducing matter by hand at the level of the potential equation.

We now generalize the analysis to the cosmological drift configuration

$$a \neq 1, \quad b \neq 0,$$

with

$$W^i = b(t, x)x^i.$$

The spatial derivative becomes

$$\partial_j W^i = b\delta_j^i + x^i \partial_j b.$$

Taking the divergence gives

$$\partial_i W^i = 3b + x^i \partial_i b.$$

Thus the torsion trace is

$$T_0 = \partial_i W^i,$$

or explicitly,

$$T_0 = 3b + x^i \partial_i b.$$

Differentiating once more yields

$$\partial_j T_0 = 3\partial_j b + \partial_j(x^i \partial_i b).$$

Expanding the derivative gives

$$\partial_j T_0 = 3\partial_j b + \delta_j^i \partial_i b + x^i \partial_j \partial_i b,$$

and therefore

$$\partial_j T_0 = 4\partial_j b + x^i \partial_j \partial_i b.$$

The quadratic derivative term becomes

$$(\partial_i W^j)(\partial_j W_i) = (b\delta_i^j + x^j \partial_i b)(b\delta_j^i + x^i \partial_j b).$$

Expanding all terms gives

$$= b^2 \delta_i^j \delta_j^i + 2bx^i \partial_i b + x^j x^i (\partial_i b)(\partial_j b).$$

Thus,

$$= 3b^2 + 2bx^i \partial_i b + x^i x^j \partial_i b \partial_j b.$$

The transport term evaluates to

$$W^j \partial_j T_0 = bx^j (4\partial_j b + x^i \partial_j \partial_i b),$$

which becomes

$$= 4bx^j \partial_j b + bx^j x^i \partial_j \partial_i b.$$

Combining all contributions yields

$$\nabla^2 \Phi = 3b^2 + 2bx^i \partial_i b + x^i x^j \partial_i b \partial_j b + 4bx^j \partial_j b + bx^i x^j \partial_i \partial_j b.$$

Grouping similar terms gives

$$\nabla^2 \Phi = 3b^2 + 6bx^i \partial_i b + x^i x^j \partial_i b \partial_j b + bx^i x^j \partial_i \partial_j b.$$

We now include cosmological expansion through

$$a = a(t),$$

with Hubble parameter

$$H = \frac{\dot{a}}{a}.$$

The time derivative of the drift field is

$$\partial_t W^i = \dot{b} x^i.$$

The mixed torsion tensor component is

$$T_{0j}^i = \frac{1}{a} (\dot{a} \delta_j^i - \partial_j W^i).$$

The torsion trace then becomes

$$T_0 = \frac{1}{a} (3\dot{a} - \partial_i W^i).$$

Substituting the divergence gives

$$T_0 = \frac{1}{a} (3\dot{a} - 3b - x^i \partial_i b),$$

or equivalently,

$$T_0 = 3H - \frac{1}{a} (3b + x^i \partial_i b).$$

Taking another derivative yields

$$\partial_j T_0 = -\frac{1}{a} (3\partial_j b + \partial_j (x^i \partial_i b)) + 3\partial_j H.$$

Since the cosmological background is homogeneous,

$$\partial_j H = 0.$$

Therefore,

$$\partial_j T_0 = -\frac{1}{a} (4\partial_j b + x^i \partial_j \partial_i b).$$

The transport contribution becomes

$$W^j \partial_j T_0 = -\frac{b}{a} x^j (4\partial_j b + x^i \partial_j \partial_i b).$$

Finally, the generalized cosmological Poisson-type structure takes the form

$$\nabla^2 \Phi = (\partial_i W^j)(\partial_j W_i) + W^j \partial_j T_0 + 3H \left(H - \frac{b}{a} \right).$$

Hence,

$$\boxed{\nabla^2 \Phi = (\partial_i W^j)(\partial_j W_i) + W^j \partial_j (\partial_i W^i) + 3 \frac{\dot{a}}{a} \left(\frac{\dot{a}}{a} - b \right)}$$

is obtained.

This equation represents a generalized Poisson structure sourced entirely by the drift geometry and torsion sector. The first term measures the local shear-like deformation of the drift field, the second term describes the transport of the torsion trace along the flow, and the final term encodes the interaction between cosmological expansion and the homogeneous drift background.

13.3 Local Vacuum Solution

In this section, we derive the local vacuum solution generated by the drift sector and show that it reproduces the Schwarzschild geometry in Painlevé–Gullstrand form.

We begin with the tetrad

$$e^0 = dt, \quad e^i = dx^i + W^i dt.$$

We consider the static local vacuum regime, where cosmological expansion is negligible. Therefore,

$$a \rightarrow 1, \quad H \rightarrow 0, \quad \partial_t W^i = 0.$$

The spacetime metric is constructed from the tetrad through

$$ds^2 = -(e^0)^2 + \delta_{ij} e^i e^j.$$

Substituting the tetrad components gives

$$ds^2 = -dt^2 + \delta_{ij} (dx^i + W^i dt)(dx^j + W^j dt).$$

Expanding the expression yields

$$ds^2 = -(1 - W^2)dt^2 + 2W_i dx^i dt + \delta_{ij} dx^i dx^j,$$

where

$$W^2 := \delta_{ij} W^i W^j.$$

To impose spherical symmetry, we choose a purely radial drift configuration:

$$W^i = W_r(r)n^i, \quad n^i = \frac{x^i}{r}.$$

Then

$$W_i dx^i = W_r n_i dx^i.$$

Using the identity

$$n_i dx^i = dr,$$

we obtain

$$W_i dx^i = W_r dr.$$

The spatial Euclidean metric in spherical coordinates is

$$\delta_{ij} dx^i dx^j = dr^2 + r^2 d\Omega^2.$$

Therefore, the metric becomes

$$ds^2 = -(1 - W_r^2) dt^2 + 2W_r dr dt + dr^2 + r^2 d\Omega^2.$$

This metric is already in Painlevé–Gullstrand form, with the drift field playing the role of the radial flow velocity.

We now define the effective drift potential as

$$\Phi_{\text{drift}} := -\frac{1}{2}W_r^2.$$

To recover the Newtonian weak-field limit, we identify this with the standard Newtonian gravitational potential

$$\Phi_{\text{N}}(r) = -\frac{GM}{r}.$$

Thus,

$$\Phi_{\text{drift}} = \Phi_{\text{N}}.$$

Substituting the definitions gives

$$-\frac{1}{2}W_r^2 = -\frac{GM}{r},$$

which immediately implies

$$W_r^2 = \frac{2GM}{r}.$$

Hence,

$$W_r = \pm \sqrt{\frac{2GM}{r}}.$$

The physically relevant branch corresponds to inward radial flow, so we choose

$$W_r < 0.$$

Therefore,

$$W_r(r) = -\sqrt{\frac{2GM}{r}}.$$

Substituting

$$W_r^2 = \frac{2GM}{r}$$

into the metric yields

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dt^2 - 2\sqrt{\frac{2GM}{r}} dr dt + dr^2 + r^2 d\Omega^2.$$

Thus the local vacuum geometry becomes

$$\boxed{ds^2 = - \left(1 - \frac{2GM}{r}\right) dt^2 - 2\sqrt{\frac{2GM}{r}} dr dt + dr^2 + r^2 d\Omega^2}$$

which is precisely the Schwarzschild spacetime written in Painlevé–Gullstrand coordinates.

To verify the equivalence explicitly, define the new time coordinate

$$dT = dt + \frac{\sqrt{2GM/r}}{1 - 2GM/r} dr.$$

Equivalently,

$$dt = dT - \frac{\sqrt{2GM/r}}{1 - 2GM/r} dr.$$

Substituting this transformation into the metric diagonalizes the line element and gives

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dT^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2.$$

Therefore,

$$ds^2 = - \left(1 - \frac{2GM}{r}\right) dT^2 + \left(1 - \frac{2GM}{r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

is recovered exactly.

This result is highly important. It shows that the drift tetrad naturally reproduces the Schwarzschild vacuum solution without introducing curvature by hand. Instead, the geometry emerges directly from the radial torsional flow encoded in the drift field $W_r(r)$.

In this interpretation, the Schwarzschild geometry may be viewed as a stationary inward drift configuration of spacetime itself. The Painlevé–Gullstrand flow velocity is therefore not merely a coordinate artifact, but arises naturally from the underlying torsional drift structure of the theory.

13.4 Derivation of the Critical Radius

In this section, we derive the characteristic critical radius at which local Newtonian attraction and cosmological drift expansion become comparable.

The effective expansion rate in the presence of the drift sector was previously defined as

$$\bar{\mathcal{H}}_{\text{eff}} = H - \frac{\bar{b}}{a},$$

where $H = \dot{a}/a$ is the standard Hubble parameter and $\bar{b}$ denotes the homogeneous background component of the drift field.

The corresponding effective cosmological acceleration is defined by

$$\mathcal{A}_{\text{cos}} = \dot{\bar{\mathcal{H}}}_{\text{eff}} + \bar{\mathcal{H}}_{\text{eff}}^2.$$

Substituting the effective Hubble parameter gives

$$\mathcal{A}_{\text{cos}} = \dot{H} - \frac{d}{dt} \left(\frac{\bar{b}}{a} \right) + \left(H - \frac{\bar{b}}{a} \right)^2.$$

The time derivative of the drift contribution is

$$\frac{d}{dt} \left(\frac{\bar{b}}{a} \right) = \frac{\dot{\bar{b}}}{a} - H \frac{\bar{b}}{a}.$$

Therefore, the effective cosmological acceleration becomes

$$\mathcal{A}_{\text{cos}} = \dot{H} - \frac{\dot{\bar{b}}}{a} + H \frac{\bar{b}}{a} + \left(H - \frac{\bar{b}}{a} \right)^2.$$

This quantity determines the large-scale repulsive or expansive effect generated jointly by cosmic expansion and the drift sector.

Now consider the radial motion of a test particle around a localized gravitating source of mass M . The effective radial equation is

$$\ddot{R} = -\frac{GM}{R^2} + \mathcal{A}_{\text{cos}} R.$$

The first term is the usual Newtonian attraction, while the second term represents the cosmological drift contribution.

The critical radius R_* is defined as the scale at which these two effects exactly balance each other:

$$\frac{GM}{R_*^2} = \mathcal{A}_{\text{cos}} R_*.$$

Multiplying both sides by R_*^2 yields

$$R_*^3 = \frac{GM}{\mathcal{A}_{\text{cos}}}.$$

Thus the critical radius is

$$R_* = \left(\frac{GM}{\mathcal{A}_{\text{cos}}} \right)^{1/3}$$

This scale separates locally bound dynamics from cosmological drift-dominated expansion.

To better understand the role of the drift background, consider the proportional ansatz

$$\frac{\bar{b}}{a} = \lambda H,$$

where λ is a dimensionless parameter characterizing the strength of the homogeneous drift sector relative to the Hubble expansion.

The effective expansion rate then becomes

$$\bar{\mathcal{H}}_{\text{eff}} = (1 - \lambda)H.$$

Consequently, the cosmological acceleration takes the form

$$\mathcal{A}_{\text{cos}} = (1 - \lambda)\dot{H} + (1 - \lambda)^2 H^2.$$

This shows explicitly that the drift background modifies both the effective expansion rate and the associated cosmic acceleration.

In the de Sitter limit, the expansion rate becomes constant:

$$\dot{H} = 0, \quad H^2 = \frac{\Lambda c^2}{3}.$$

Substituting these relations into the critical-radius formula gives

$$R_* = \left(\frac{3GM}{\Lambda c^2} \right)^{1/3}.$$

Therefore,

$$R_* = \left(\frac{3GM}{\Lambda c^2} \right)^{1/3}$$

is recovered as the de Sitter critical scale.

This result demonstrates that the present drift framework naturally reproduces the standard cosmological turnaround scale while simultaneously allowing modifications through the drift contribution $\bar{b}$. The critical radius is therefore dynamically controlled not only by the cosmological constant but also by the torsional drift sector encoded in the effective expansion rate.

14 The Detailed Derivations of Baryonic Tully Fisher Relation

14.1 Derivation of the Baryonic Tully–Fisher Relation from Drift Dynamics

In this section, we show how the baryonic Tully–Fisher relation naturally emerges from the drift sector of the theory.

Inside the baryon-dominated region of a galaxy, the Newtonian acceleration produced by baryonic matter is approximately

$$g_{\text{bar}}(r) \simeq \frac{GM_{\text{bar}}(r)}{r^2}.$$

For circular motion, the centripetal acceleration satisfies

$$\frac{v_f^2}{r} = g(r),$$

which implies

$$v_f^2 = r g(r).$$

In the previous section, the drift-induced acceleration was found to behave as

$$g_W(r) = W(r)W'(r).$$

Therefore,

$$v_f^2 = rW(r)W'(r).$$

In the flat rotation regime, one obtains

$$W(r)W'(r) = \frac{C}{r},$$

so that the drift acceleration behaves as

$$g_W(r) = \frac{C}{r}.$$

We now consider the transition region between the Newtonian and drift-dominated regimes. In this region, the baryonic and drift accelerations are assumed to be of the same order:

$$g_{\text{bar}}(r_*) \sim g_W(r_*).$$

Thus,

$$\frac{GM_{\text{bar}}}{r_*^2} \sim \frac{C}{r_*},$$

which gives

$$C = \frac{GM_{\text{bar}}}{r_*}.$$

At the same transition scale, the drift acceleration is also assumed to be of the order of the MOND acceleration scale:

$$g_W(r_*) \sim a_0.$$

Hence,

$$\frac{C}{r_*} \sim a_0,$$

leading to

$$C \sim a_0 r_*.$$

Therefore, two independent expressions for C are obtained:

$$C = \frac{GM_{\text{bar}}}{r_*}, \quad C = a_0 r_*.$$

Combining these relations yields

$$C^2 = \frac{GM_{\text{bar}}}{r_*}(a_0 r_*),$$

and consequently

$$C^2 = GM_{\text{bar}} a_0.$$

From the previous section,

$$v_f^2 = C,$$

so that

$$v_f^4 = C^2.$$

Therefore,

$$\boxed{v_f^4 = GM_{\text{bar}} a_0}$$

is obtained directly.

This is precisely the baryonic Tully–Fisher relation. In the present framework, it emerges naturally from the asymptotic drift geometry.

14.2 Generation of Flat Rotation Curves from the Drift Field

In this section, we explicitly show how the drift field produces flat galactic rotation curves.

In the previous analysis, the effective potential generated by the drift sector was found to be

$$\Phi(r) = -\frac{1}{2}W^2(r).$$

By definition, the radial acceleration is

$$a(r) = -\frac{d\Phi}{dr},$$

which gives

$$a(r) = W(r)W'(r).$$

For circular motion, the centripetal acceleration is

$$a(r) = \frac{v^2(r)}{r}.$$

Therefore,

$$\frac{v^2(r)}{r} = W(r)W'(r),$$

and thus

$$v^2(r) = rW(r)W'(r).$$

In the outer galactic region, observed rotation curves are approximately flat, meaning that the velocity becomes nearly constant:

$$v^2(r) = v_f^2.$$

Hence the drift field must satisfy

$$rW(r)W'(r) = v_f^2,$$

or equivalently,

$$W(r)W'(r) = \frac{v_f^2}{r}.$$

Defining

$$C \equiv v_f^2,$$

one obtains

$$W(r)W'(r) = \frac{C}{r}.$$

This differential equation can be solved directly. First rewrite it as

$$W \frac{dW}{dr} = \frac{C}{r}.$$

Integrating both sides gives

$$\int_{W_0}^W W dW = \int_{r_0}^r \frac{C}{r} dr.$$

Thus,

$$\frac{W^2}{2} - \frac{W_0^2}{2} = C \ln r - C \ln r_0,$$

or equivalently,

$$W^2(r) = W_0^2 + 2C \ln \left(\frac{r}{r_0} \right).$$

Substituting $C = v_f^2$ yields

$$W^2(r) = W_0^2 + 2v_f^2 \ln \left(\frac{r}{r_0} \right)$$

Therefore, the drift field exhibits logarithmic growth in the outer galactic region, while the resulting acceleration behaves as

$$a(r) = W(r)W'(r) \sim \frac{1}{r}.$$

This asymptotic behavior directly generates flat galactic rotation curves.

14.3 Consistency Check of the MOND-like Regime

In this section, we show that the solution

$$W(r)W'(r) = \frac{C}{r}$$

is asymptotically consistent with the drift field equation.

The drift equation obtained in the matter sector is

$$\beta \left(W'' + \frac{2W'}{r} - \frac{2W}{r^2} \right) = W(1+W) \left(\frac{\beta \left(W' + \frac{2W}{r} \right)^2 - \frac{2W}{r} \left(2W' + \frac{W}{r} \right)}{2(2W^2 - 1)} \right).$$

In the MOND-like regime, the desired asymptotic behavior is

$$W(r)W'(r) = \frac{C}{r},$$

which implies

$$W' = \frac{C}{rW}.$$

Therefore,

$$(W')^2 = \frac{C^2}{r^2 W^2}.$$

We now compute the second derivative:

$$W'' = \left(\frac{C}{rW} \right)'.$$

This gives

$$W'' = C \left(\frac{1}{rW} \right)'.$$

Using

$$\frac{d}{dr}(rW)^{-1} = -\frac{(rW)'}{(rW)^2} = -\frac{W + rW'}{r^2W^2},$$

one obtains

$$W'' = -C \frac{W + rW'}{r^2W^2}.$$

Substituting

$$W' = \frac{C}{rW},$$

gives

$$W'' = -\frac{C}{r^2W} - \frac{C^2}{r^2W^3}.$$

Now insert this into

$$W'' + \frac{2W'}{r} - \frac{2W}{r^2}.$$

One finds

$$-\frac{C}{r^2W} - \frac{C^2}{r^2W^3} + \frac{2C}{r^2W} - \frac{2W}{r^2}.$$

Combining terms yields

$$W'' + \frac{2W'}{r} - \frac{2W}{r^2} = \frac{1}{r^2} \left(\frac{C}{W} - \frac{C^2}{W^3} - 2W \right).$$

We now examine the right-hand side of the field equation.

Using

$$(W')^2 = \frac{C^2}{r^2W^2}$$

and

$$WW' = \frac{C}{r},$$

the right-hand side becomes

$$\frac{W(1+W)}{2(2W^2-1)} \left(\frac{\beta C^2}{r^2W^2} + \frac{4(\beta-1)C}{r^2} + \frac{(4\beta-2)W^2}{r^2} \right).$$

Factoring out the common scale gives

$$\frac{1}{r^2} \frac{W(1+W)}{2(2W^2-1)} \left(\frac{\beta C^2}{W^2} + 4(\beta-1)C + (4\beta-2)W^2 \right).$$

Therefore, both sides of the field equation scale asymptotically as

$$\frac{1}{r^2}.$$

This result is highly important. If the two sides exhibited different asymptotic scalings, the equation would become inconsistent in the large- r limit.

Hence the solution

$$W(r)W'(r) = \frac{C}{r}$$

is asymptotically consistent with the drift field equation.

Physically, this demonstrates that the drift sector naturally generates the low-acceleration behavior

$$a(r) \sim \frac{1}{r},$$

which is required for flat galactic rotation curves.

Thus, both flat rotation curves and the baryonic Tully–Fisher relation emerge directly from the structure of the torsion-based drift geometry itself, rather than from a phenomenologically imposed interpolation function.

15 Further Physical Analysis

15.1 Degree of Freedom Analysis

In this appendix, we analyze the dynamical structure of the vector mode introduced through the isotropic ansatz $W^i = b(t, \mathbf{x})x^i$. The goal is to determine whether the field b propagates an independent degree of freedom.

Action and Background Definitions

We begin with the modified teleparallel action

$$S = \frac{1}{2\kappa} \int d^4x e (T + \beta T_\mu T^\mu) \quad (271)$$

where T is the torsion scalar and T_μ is the torsion vector. The parameter β controls the strength of the vector-torsion correction.

For the isotropic vector configuration we define

$$W^i = b(t, \mathbf{x})x^i \quad (272)$$

where $b(t, \mathbf{x})$ is the scalar function associated with the vector mode.

To simplify the notation we introduce the combinations

$$X = H - \frac{\dot{b}}{a} \quad (273)$$

and

$$Y = \frac{x^i \partial_i b}{a} \quad (274)$$

where $H = \dot{a}/a$ is the Hubble parameter. The quantity X measures the deviation from the standard FRW background, while Y contains the spatial derivative contribution of b .

Using these definitions, the gravitational Lagrangian density becomes

$$\mathcal{L}_g = \frac{ca^3}{2\kappa} [(6 - 9\beta)X^2 + (-4 + 6\beta)XY - \beta Y^2] \quad (275)$$

Absence of a Kinetic Term

A crucial observation is that the Lagrangian does not contain a time derivative of b ,

$$\dot{b} \notin \mathcal{L}_g \quad (276)$$

which immediately implies that the canonical momentum conjugate to b vanishes:

$$\pi_b := \frac{\partial \mathcal{L}_g}{\partial \dot{b}} \quad (277)$$

$$\pi_b = 0 \quad (278)$$

This defines the primary constraint

$$\Phi_1 := \pi_b \approx 0 \quad (279)$$

where the symbol ≈ 0 denotes a weak equality in the Dirac constraint formalism.

Total Hamiltonian

The total Hamiltonian is constructed by adding the primary constraint with a Lagrange multiplier u_b ,

$$H_{\text{tot}} = \int d^3x \left(\pi_b \dot{b} - \mathcal{L}_g + u_b \pi_b \right) \quad (280)$$

Since $\pi_b = 0$, this reduces to

$$H_{\text{tot}} = \int d^3x (-\mathcal{L}_g) \quad (281)$$

Secondary Constraint

Consistency of the primary constraint requires

$$\dot{\Phi}_1 = \{\pi_b, H_{\text{tot}}\} \quad (282)$$

or equivalently

$$\dot{\pi}_b = -\frac{\delta H_{\text{tot}}}{\delta b} \quad (283)$$

which gives

$$\dot{\pi}_b = \frac{\delta \mathcal{L}_g}{\delta b} \quad (284)$$

The preservation condition

$$\dot{\pi}_b \approx 0 \quad (285)$$

therefore generates the secondary constraint

$$\Phi_2 := \frac{\partial \mathcal{L}_g}{\partial b} - \partial_i \left(\frac{\partial \mathcal{L}_g}{\partial (\partial_i b)} \right) \approx 0 \quad (286)$$

which is simply the Euler–Lagrange equation for b .

Evaluation of the Constraint

Using the definitions of X and Y ,

$$\frac{\partial X}{\partial b} = -\frac{1}{a} \quad (287)$$

$$\frac{\partial Y}{\partial b} = 0 \quad (288)$$

$$\frac{\partial Y}{\partial (\partial_i b)} = \frac{x^i}{a} \quad (289)$$

we compute

$$\frac{\partial \mathcal{L}_g}{\partial b} = \frac{ca^3}{2\kappa} \left[2(6 - 9\beta)X \frac{\partial X}{\partial b} + (-4 + 6\beta)Y \frac{\partial X}{\partial b} \right] \quad (290)$$

which simplifies to

$$\frac{\partial \mathcal{L}_g}{\partial b} = -\frac{ca^2}{2\kappa} [2(6 - 9\beta)X + (-4 + 6\beta)Y] \quad (291)$$

Similarly,

$$\frac{\partial \mathcal{L}_g}{\partial(\partial_i b)} = \frac{ca^3}{2\kappa} \left[(-4 + 6\beta)X \frac{\partial Y}{\partial(\partial_i b)} - 2\beta Y \frac{\partial Y}{\partial(\partial_i b)} \right] \quad (292)$$

leading to

$$\frac{\partial \mathcal{L}_g}{\partial(\partial_i b)} = \frac{ca^2}{2\kappa} x^i [(-4 + 6\beta)X - 2\beta Y] \quad (293)$$

Hence the secondary constraint becomes

$$\Phi_2 = -\frac{ca^2}{2\kappa} [2(6 - 9\beta)X + (-4 + 6\beta)Y] - \partial_i \left[\frac{ca^2}{2\kappa} x^i ((-4 + 6\beta)X - 2\beta Y) \right] \quad (294)$$

with

$$\Phi_2 \approx 0 \quad (295)$$

Constraint Algebra

We now classify the constraints using Poisson brackets.

The primary constraint trivially satisfies

$$\{\Phi_1(\mathbf{x}), \Phi_1(\mathbf{y})\} = 0 \quad (296)$$

while

$$\{\Phi_1(\mathbf{x}), \Phi_2(\mathbf{y})\} = -\frac{\delta \Phi_2(\mathbf{y})}{\delta b(\mathbf{x})} \quad (297)$$

Using

$$\frac{\delta X}{\delta b} = -\frac{1}{a} \delta(\mathbf{x} - \mathbf{y}) \quad (298)$$

and

$$\frac{\delta Y}{\delta b} = \frac{x^i \partial_i}{a} \delta(\mathbf{x} - \mathbf{y}) \quad (299)$$

we find that

$$\frac{\delta \Phi_2}{\delta b} \neq 0 \quad (300)$$

and therefore

$$\{\Phi_1, \Phi_2\} \neq 0 \quad (301)$$

This means that the constraints form a second-class pair:

$$\Phi_1, \Phi_2 \text{ second-class} \quad (302)$$

Degree of Freedom Counting

The phase space associated with the variable b contains

$$N_{\text{phase}} = 2 \quad (303)$$

corresponding to the pair (b, π_b) .

There are no first-class constraints,

$$N_{\text{first-class}} = 0 \quad (304)$$

and two second-class constraints,

$$N_{\text{second-class}} = 2 \quad (305)$$

Using the standard Dirac counting formula,

$$N = \frac{1}{2} (N_{\text{phase}} - 2N_{\text{first-class}} - N_{\text{second-class}}) \quad (306)$$

we obtain

$$N = \frac{1}{2} (2 - 0 - 2) = 0 \quad (307)$$

Therefore the field b does not propagate any independent physical degree of freedom.

Degeneracy of the Kinetic Matrix

The absence of propagation is also confirmed by the degeneracy of the kinetic sector:

$$\frac{\partial^2 \mathcal{L}_g}{\partial \dot{b}^2} = 0 \quad (308)$$

which implies that the kinetic matrix eigenvalue vanishes,

$$K_b = 0 \quad (309)$$

As a consequence, no standard dispersion relation exists for the mode,

$$\omega_b^2 \text{ undefined} \quad (310)$$

and the final result is

$$N = 0 \quad (311)$$

Thus, the isotropic vector configuration introduces no additional propagating scalar degree of freedom.

15.2 Tensor Sector

In this appendix we briefly analyze the tensor perturbations of the theory in order to determine whether the additional torsion-vector term

$$\beta T_\mu T^\mu \quad (312)$$

modifies the propagation of gravitational waves.

We consider the standard transverse-traceless (TT) tensor perturbation around the flat FRW background,

$$e^i_j = a(t) \left(\delta^i_j + \frac{1}{2} h^i_j \right) \quad (313)$$

with

$$\partial_i h^{ij} = 0, \quad h^i_i = 0 \quad (314)$$

The torsion vector is defined as

$$T_\mu = T^\nu_{\nu\mu} \quad (315)$$

Expanding the torsion tensor to linear order in h_{ij} , one finds

$$\delta T_\mu^{TT} = 0 \quad (316)$$

Therefore the quadratic tensor contribution of the new invariant vanishes:

$$(T_\mu T^\mu)^{(2)}_{TT} = 0 \quad (317)$$

and consequently

$$S_{\beta,TT}^{(2)} = 0 \quad (318)$$

Hence the tensor sector receives no correction from the $\beta T_\mu T^\mu$ term, and the quadratic

action reduces to the standard TEGR/GR result,

$$S_{TT}^{(2)} = \frac{1}{8\kappa} \int dt d^3x a^3 \left[\dot{h}_{ij} \dot{h}^{ij} - \frac{1}{a^2} \partial_k h_{ij} \partial^k h^{ij} \right] \quad (319)$$

The resulting equation of motion is

$$\ddot{h}_{ij} + 3H\dot{h}_{ij} - \frac{1}{a^2} \nabla^2 h_{ij} = 0 \quad (320)$$

which is identical to General Relativity. For plane-wave solutions,

$$h_{ij} \sim e^{i(\mathbf{k} \cdot \mathbf{x} - \omega t)} \quad (321)$$

the dispersion relation becomes

$$\omega^2 = k^2 \quad (322)$$

implying

$$c_T^2 = 1 \quad (323)$$

Therefore the theory propagates the same two tensor polarizations as GR, with no modification to the gravitational-wave speed.